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Inventor(s)	Crain; Stephen G. et al.

Systems and methods for engine start

Abstract

Systems and methods for starting and restarting an engine of a vehicle are disclosed. Power systems for an engine are disclosed.

Inventors: Crain; Stephen G. (Wyoming, MN), Rhodes; Trevor F. (Badger, MN), Kelay; Yassin M. (Sugiez, CH), Buehler; David F. (Seedorf, CH), Hoffet; Markus (Kappelen, CH), Buchwitz; James H. (Strathcona, MN), Hartung; Nicholas A. M. (Medina, MN), Jaeger; Ricky G. (Osceola, WI), Danielson; Ronnie R. (Roseau, MN), Blake; Dallas J. (Roseau, MN), Kohler; Beat R. (Kirchberg, CH), Gasdia; Philippe (Rothrist, CH), Tharaldson; Joseph D. (Taylor Falls, MN), Hebbard; Garrett J. (Chisago City, MN), Bilek; Andreas H. (Chisago City, MN), Zimney; Derek (Roseau, MN)

Applicant: Polaris Industries Inc. (Medina, MN)

Family ID: 1000008819828

Assignee: Polaris Industries Inc. (Medina, MN)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4930467	12/1989	Masuda	N/A	N/A
6176218	12/2000	Beechie	123/339.22	F02D 31/008
7054738	12/2005	Stotsky	701/115	F02D 41/1402
7324888	12/2007	Stotsky	701/101	F02D 41/1498
9845782	12/2016	Bernier	N/A	F02N 19/005
10883467	12/2020	Lebreux	N/A	N/A
10900455	12/2020	Bernier	N/A	F02B 75/04
11352996	12/2021	Bernier	N/A	F02N 11/08
11739720	12/2022	Bernier	123/179.3	F02N 11/08
2006/0169237	12/2005	Yang	123/179.5	F02N 11/0866
2008/0111524	12/2007	Rydman	320/166	F02N 11/0866
2009/0020092	12/2008	Kishibata	N/A	N/A
2009/0061705	12/2008	Wiatrowski	440/85	F02N 11/0862
2014/0375066	12/2013	Yang	290/38R	F02N 11/0862
2014/0375067	12/2013	Yang	290/38R	B60R 16/03
2015/0083079	12/2014	Li	123/350	F02N 11/08
2018/0347534	12/2017	Lebreux	N/A	F02N 11/0803
2019/0153988	12/2018	Lebreux	N/A	F02N 11/06
2019/0153989	12/2018	Lebreux	N/A	F02N 11/06
2019/0153990	12/2018	Lebreux	N/A	F02N 11/06
2020/0056559	12/2019	Lebreux	N/A	F02N 11/0866
2020/0080503	12/2019	Lebreux	N/A	F02D 41/062
2020/0080504	12/2019	Lebreux	N/A	F02N 11/06

2020/0080505	12/2019	Lebreux	N/A	F02B 33/04
2020/0136414	12/2019	Patsos	N/A	H02J 7/342
2020/0347812	12/2019	Hashizume	N/A	F02N 11/0862
2021/0062774	12/2020	Lebreux	N/A	F02N 11/0866
2022/0364520	12/2021	Lebreux	N/A	F02P 5/045
2024/0084752	12/2023	Bernier	N/A	F02B 61/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017/081664	12/2016	WO	N/A
2018/150232	12/2017	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion received for PCT Patent Application No.

PCT/US2022/076564, mailed on Feb. 13, 2023, 14 pages. cited by applicant

International Preliminary Report on Patentability issued by the International Bureau of WIPO, dated Mar. 5, 2024, for International Patent Application No. PCT/US2022/076564; 8 pages. cited by applicant

Primary Examiner: Zaleskas; John M

Attorney, Agent or Firm: Merchant & Gould P.C.

Background/Summary

CROSS-REFERENCE (1) The present application claims the benefit of U.S. Provisional Patent Application No. 63/244,897, filed Sep. 16, 2021, titled SYSTEMS AND METHODS FOR ENGINE START, the entire disclosure of which is expressly incorporated by reference herein.

TECHNICAL FIELD

(1) The present disclosure relates generally to an engine starting system and, more particularly, to an engine starting system having an electric start control.

BACKGROUND

(2) Generally, to start or restart an engine of a recreational vehicles, such as snowmobiles, a user of a vehicle may pull on a rope of a recoil starter to cause a crankshaft of the engine to turn. The pull needs to be fast enough to turn the crankshaft to reach a threshold speed to manually start the engine. However, the manual start of the engine by pulling the rope at the start or every restart of the engine during the ride may cause fatigue in the user.

SUMMARY

(3) As set forth above, embodiments provided herein relate to engine starting system of a recreational vehicle. Exemplary embodiments include but are not limited to the following examples.

(4) In one aspect, a method for controlling different starting modes of an internal combustion engine of a vehicle is provided. The method includes determining whether a power level of a power source is above a first predetermined threshold, determining, in response to determining that the power level is above the first predetermined threshold, that a full-starting mode is available, determining, in response to determining that the power level is not above the first predetermined threshold, whether the power level is below a second predetermined threshold, determining, in response to determining that the power level is below the second predetermined threshold, that a manual starting mode is available, determining, in response to determining that the power level is

not below the second predetermined threshold, that a pull-assist starting mode is available, and transmitting a notification to a user indicating an available starting mode.

(5) In some embodiments, the method may further include receiving a user request to start an engine prior to determining the power level, wherein the user request is received via a user input sensor of the vehicle.

(6) In some embodiments, the method may further include receiving a user request to start an engine subsequent to determining the power level, wherein the user request is received via a user input sensor of the vehicle.

(7) In some embodiments, the user input sensor may include a button, a pull recoil handle sensor, a pull force sensor, and/or a recoil rotation sensor.

(8) In some embodiments, the full-starting mode may support electric start or restart of the engine by using the power source to deliver electrical power to an engine control module (ECM).

(9) In some embodiments, the manual starting mode may support manual pull start via a rope of a recoil starter.

(10) In some embodiments, the pull-assist starting mode may support electric start with manual pull assist.

(11) In some embodiments, in the pull-assist starting mode, when a rope of a recoil starter is pulled by the user, current may be driven into a stator to apply cranking assist torque to a crankshaft to turn the engine.

(12) In some embodiments, the notification may be displayed on a display screen of the vehicle.

(13) In some embodiments, the notification may be a text notification and/or a light indicator with a blinking pattern and/or color that corresponds to a different starting mode.

(14) In another aspect, a hybrid battery system for starting and restarting an engine of a vehicle is provided. The hybrid battery system includes a DC/DC converter, a supercapacitor system configured to provide electric power to the engine during engine starting at low temperature and be recharged when the engine is running, and a lithium-ion battery removably connected to the vehicle and configured to pre-charge the supercapacitor system when the engine is not running.

(15) In other aspect, a method for restarting an engine in a reverse direction is provided. The method includes receiving, by an engine control unit, a user request to reverse engine rotation while engine is running, determining, by the engine control unit, whether one or more parameters satisfy predefined condition to perform reverse engine rotation, performing, in response to determining that the one or more parameters satisfy reverse engine rotation requirements and by the engine control unit, an engine reverse process by switching a starter-generator controller to a starter mode, applying, by the starter-generator controller, braking torque to a crankshaft of the vehicle by sending excess power to a power source to decelerate a speed of the crankshaft, developing, by the starter-generator controller, torque in an opposite direction of a current rotation of the crankshaft by using energy from the power source, and switching, in response to determining that the crankshaft is rotating in the opposite direction, by the engine control unit, the starter-generator controller back to a voltage regulator mode and recharge the power source.

(16) In some embodiments, receiving the user request to reverse engine rotation while engine is running may include receiving the user request via a user input of the vehicle, wherein the user input includes a pushbutton of the vehicle or a selection on a touchscreen display that is communicatively coupled to the vehicle.

(17) In some embodiments, performing the engine reverse process by switching a starter-generator controller to a starter mode may include disabling, by the engine control unit, spark and fuel to the engine, transmitting, by the engine control unit, a signal to the starter-generator controller to switch the starter-generator controller to the starter mode.

(18) In some embodiments, the method may further include determining, by the starter-generator controller, if the power source are fully charged, and shunting, in response to determining that the power source are fully charged and by the starter-generator controller, a stator to provide the

braking torque to the crankshaft.

(19) In some embodiments, developing torque in the opposite direction of the current rotation of the crankshaft may include determining if a speed of the crankshaft is below a predefined threshold, and developing, in response determining that the speed of the crankshaft is below a predefined threshold, torque in the opposite direction of the current rotation of the crankshaft by using energy from the power source.

(20) In some embodiments, switching the starter-generator controller back to the voltage regulator mode may include enabling, by the engine control unit, the spark and fuel to restart the engine.

(21) In another aspect, an electric reverse control system for restarting an engine in a reverse direction is provided. The electric reverse control system includes a power source, a reverse control user input configured to receive a user request to reverse engine rotation, an engine control module (ECM), and a starter-generator controller having a starter mode and a voltage regulator mode. The ECM is configured to receive the user request to reverse engine rotation while engine is running, determine whether one or more parameters satisfy predefined condition to perform reverse engine rotation, perform, in response to determination that the one or more parameters satisfy reverse engine rotation requirements, an engine reverse process by switching a starter-generator controller to a starter mode, control operations of the starter-generator controller. The starter-generator controller is configured to apply braking torque to a crankshaft of the vehicle by sending excess power to the power source to decelerate a speed of the crankshaft, and develop torque in an opposite direction of a current rotation of the crankshaft by using energy from the power source.

(22) In some embodiments, the user input may include a pushbutton of the vehicle or a selection on a touchscreen display that is communicatively coupled to the vehicle.

(23) In some embodiments, to perform the engine reverse process by switching a starter-generator controller to a starter mode may include to disable spark and fuel to the engine, and transmit a signal to the starter-generator controller to switch the starter-generator controller to the starter mode.

(24) In some embodiments, the starter-generator controller may be further configured to determine if the power source are fully charged, and shunt, in response to determination that the power source are fully charged, a stator to provide the braking torque to the crankshaft.

(25) In some embodiments, to develop torque in the opposite direction of the current rotation of the crankshaft may include to determine if a speed of the crankshaft is below a predefined threshold, and develop, in response determination that the speed of the crankshaft is below a predefined threshold, torque in the opposite direction of the current rotation of the crankshaft by using energy from the power source.

(26) In some embodiments, to switch the starter-generator controller back to the voltage regulator mode may include to enable the spark and fuel to restart the engine.

(27) In other aspect, an engine starting system for starting and restarting an engine of a vehicle is provided. The engine starting system includes a DC/DC converter, an integrated starter generator (ISG) configured to recharge the lithium-ion capacitors when the engine is running, and lithium-ion capacitors configured to provide power to start and restart the engine.

(28) In another aspect, a starter-generator system for providing electric start of an engine of a vehicle is provided. The starter-generator system includes an integrated starter-generator (ISG), a power source module including a power source and a bidirectional DC/DC voltage converter, wherein the power source is configured to supply power during engine starting and is recharged when the engine is running and the bidirectional DC/DC voltage converter is configured to provide charging of the power source, an energy storage configured to supply power when engine is running, and a starter-generator (ISG) controller configured to apply power to a stator for engine starting from the power source and to take power from the stator when running to supply an electrical load of vehicle from the energy storage.

(29) In other aspect, a starter-generator system for providing electric start of an engine of a vehicle

is provided. The starter-generator system includes a removeable power source removably coupled to the vehicle, a bidirectional DC/DC converter for charging the removeable power source from a vehicle power when the engine is running, supplying power to the vehicle prior to engine starting, and supplying power to in-vehicle infotainment and vehicle functions when an ignition key is off, and an integrated starter generator (ISG) operatively coupled to the removeable power source to provide a higher voltage to increase a starting torque.

(30) In some embodiments, the removeable power source may include a port for off-board charging and/or heating the removeable power source; and a button and a plurality of LED lights. The button may be used to cause the removeable power source to display the state-of-charge by turning on one or more of the plurality of LED lights for a period of time, turn on or off a white LED to use as a flashlight, and/or turn on or off a self-heating mode to function as a hand warmer and/or preheat the removeable power source prior to be coupled to the vehicle.

(31) In another aspect, a method for supplying power for vehicle starting from a removeable power source is provided. The method includes receiving a first user input while the removable power source is in an off state, transitioning, in response to receiving the first user input, the removable power source to a wake state, determining if a power capacity exceeds a predetermined threshold for electric start, transmitting, in response to determining that the power capacity exceeds the predetermined threshold for electric start, a notification to a user indicating that the electric start is possible, receiving a second user input requesting to electric start an engine, and supplying, in response to receiving the second user input, electrical power from the removable power source to an engine control module to produce starting torque.

(32) In some embodiments, transitioning the removable power source to the wake state may include providing power from the removable power source to display on an engine control module.

(33) In some embodiments, receiving the first user input may include monitoring a key switch or a power button input while the removable power source is in an off state to determine whether the first user input has been received.

(34) In some embodiments, the method may further include transmitting, in response to determining that the power capacity does not exceed the predetermined threshold for electric start, a notification to a user indicating that the electric start is not possible.

(35) In some embodiments, receiving the second user input may include receiving the second user input via a user input sensor of the vehicle, wherein the user input sensor may include a button, a pull recoil handle sensor, a pull force sensor, and/or a recoil rotation sensor.

(36) In some embodiments, determining if the power capacity exceeds the predetermined threshold for electric start may include receiving, from the removeable power source, a maximum discharge current of the removeable power source based on a power source temperature and state-of-charge of the removeable power source, receiving, from the engine control module, an engine coolant temperature, determining the predetermined threshold based on the engine coolant temperature and time since the engine was stopped, and comparing the maximum discharge current against the predetermined threshold for the electric start.

(37) In another aspect, a method for supplying power for vehicle starting from a removeable power source is provided. The method includes determining if a vehicle is turned on and engine is not running, determining if a power capacity exceeds a predetermined threshold for electric start, transmitting, in response to determining that the power capacity exceeds the predetermined threshold for electric start, a notification to a user indicating that the electric start is possible, receiving a user input requesting to electric start an engine in reverse, and operating, in response to receiving the user input, electrical power to an integrated starter-generator to turn engine in reverse direction.

(38) While multiple embodiments are disclosed, still other embodiments of the presently disclosed subject matter will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments of the disclosed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above-mentioned and other features and advantages of this disclosure, and the manner of attaining them, will become more apparent and will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 illustrates a side view of an exemplary vehicle;

(3) FIG. 2 illustrates a system for controlling different starting modes of an internal combustion engine including a starter-generator controller, according to an example of the present disclosure;

(4) FIG. 3 is a flow diagram illustrating a computer-implemented method for controlling different starting modes of an internal combustion engine controlling different starting modes of an internal combustion engine using the starter-generator controller of FIG. 2;

(5) FIG. 4 illustrates a system for starting and restarting an engine of a vehicle including a lithium-ion (Li-Ion) battery, a DC/DC converter, and a supercapacitor system, according to an example of the present disclosure;

(6) FIG. 5 illustrates a system for restarting an engine in a reverse direction, according to an example of the present disclosure;

(7) FIGS. 6 and 7 are a flow diagram illustrating a computer-implemented method for restarting an engine in a reverse direction by controlling a direction of an engine rotation of a vehicle;

(8) FIG. 8 illustrates a system for starting and restarting an engine of a vehicle, according to an example of the present disclosure;

(9) FIG. 9 illustrates a system for providing electric start, electrically assisted-starting, and power generation, according to an example of the present disclosure;

(10) FIGS. 10, 11A, and 11B are example schematic circuit diagrams of portion of the system of FIG. 9;

(11) FIGS. 12A and 12B illustrates a removable lithium-ion battery, according to an example of the present disclosure;

(12) FIG. 13 illustrates a system for supplying power to an integrated starter-generator (ISG) controller for vehicle starting, according to an example of the present disclosure;

(13) FIG. 14 illustrates different battery operating states of the removable lithium-ion battery of FIGS. 12 and 13;

(14) FIG. 15 illustrates a system including a removable battery for supplying power to an integrated starter-generator (ISG) controller for vehicle starting, according to an example of the present disclosure;

(15) FIG. 16 is a flow diagram illustrating a computer-implemented method for electric starting an engine from a removable battery using the removable battery of FIG. 15;

(16) FIGS. 17A and 17B are an example schematic circuit diagram of a portion of the system of FIG. 15;

(17) FIG. 18 is an example schematic circuit diagram of a portion of the system of FIG. 15;

(18) FIG. 19 is a flow diagram illustrating a computer-implemented method for starting an engine of a vehicle in reverse, according to an example of the present disclosure;

(19) FIG. 20 is a flow diagram illustrating a computer-implemented method for determining a starting method of an engine, according to an example of the present disclosure;

(20) FIG. 20A is a flow diagram of a subprocess engine friction test of the flow diagram of FIG. 20, according to an example of the present disclosure;

(21) FIG. 21 is a flow diagram illustrating an alternate computer-implemented method for determining a starting method of an engine, according to an example of the present disclosure;

(22) FIG. 22 is a flow diagram illustrating a computer-implemented method for determining a controller mode for an engine, according to an example of the present disclosure;

(23) FIG. 23 is a diagram illustrating a stator and shunt diagram, according to an example of the present disclosure; and

(24) FIG. 24 is a flow diagram illustrating a computer-implemented method for determining when a stator may be shunted to a ground pin, according to an example of the present disclosure.

(25) Corresponding reference characters indicate corresponding parts throughout the several views. Although the drawings represent embodiments of the present disclosure, the drawings are not necessarily to scale, and certain features may be exaggerated in order to better illustrate and explain the present disclosure. The exemplification set out herein illustrates an embodiment of the disclosure, in one form, and such exemplifications are not to be construed as limiting the scope of the disclosure in any manner.

DETAILED DESCRIPTION OF THE DRAWINGS

(26) Various embodiments of the present invention will be described in detail with reference to the drawings, wherein like reference numerals represent like parts and assemblies throughout the several views. Reference to various embodiments does not limit the scope of the invention, which is limited only by the scope of the claims attached hereto. Additionally, any examples set forth in this specification are not intended to be limiting and merely set forth some of the many possible embodiments for the claimed invention.

(27) Referring to FIG. 1, an exemplary embodiment of a vehicle **10** is shown. The illustrated vehicle **10** is a snowmobile. However, the following disclosure is applicable to other types of vehicles, such as all-terrain vehicles, motorcycles, watercraft, utility vehicles, and golf carts. Vehicle **10** includes a controller **12**, a motor **22**, an electric start system **14**, a recoil start system **16**, an alternator **20**, a fuel source **30** (e.g., gasoline), a power source **18** (e.g., battery), a continuously variable transmission (CVT) **24**, and a variety of ground engaging members **28**. Exemplary ground engaging members include skis, endless tracks, wheels, and other suitable devices which support vehicle **10** relative to the ground. For example, as illustrated in FIG. 1, ground engaging members **28** include a track assembly **32** (e.g., an endless track belt) and a pair of front skis **34**. The track assembly **32** holds the rear of vehicle **10**, while skis **34** support the front of vehicle **10**. In addition, the track assembly **32** is operatively connected to a motor **22**.

(28) Referring to FIG. 2, a starter-generator system **100** for controlling different starting modes of an internal combustion engine is described. Generally, to start or restart the engine, a user of a vehicle may pull on a rope of a recoil starter to cause a crankshaft of the engine to turn. The pull needs to be fast enough to turn the crankshaft to reach a threshold speed to manually start the engine. However, the manual start of the engine by pulling the rope at the start or every restart of the engine during the ride may cause fatigue in the user.

(29) In the illustrative embodiment, starter-generator system **100** determines an available starting mode of the engine based at least on power source health to reduce a number of manual starts performed by the user during the ride. To do so, the starter-generator system **100** includes a starter-generator controller **110**, a power source **112**, a user input sensor **114**, a start enable indicator **116**, a stator **118**, and an engine control module (ECM) **120**. ECM **120** is operatively connected to starter generator system **100** to receive power to control performance of the internal combustion engine.

(30) Starter-generator controller **110** is configured to determine available starting mode of the vehicle based at least in part on the health or level of power source **112** (e.g., a battery level). The power source **112** may be embodied as multiple interconnected batteries or ultracapacitors. In the illustrative embodiment, the starting mode option includes a full starting mode, a pull-assist starting mode, and a manual starting mode. The full starting mode supports electric start or restart of the engine by using power source **112** to deliver electrical power to ECM **120** when power source **112**

satisfies a first characteristic (e.g., above a first predetermined threshold). The manual starting mode supports manual pull start via the rope of a recoil starter when power source **112** satisfies a second characteristic (e.g., below a second predetermined threshold).

(31) Additionally, the pull-assist starting mode supports electric start with manual pull assist when power source **112** does not satisfy the first and second characteristics (e.g., between the first and second predetermined thresholds), which is described further below. By providing the pull-assist starting mode, starter-generator system **100** may provide an alternative starting option that reduces a pull force required to start the engine when the full electric start is not available. It should be appreciated that by providing the pull-assist starting option, starter-generator controller **110** may provide an alternative starting option to reduce a pull force required to start the engine without impacting a weight of the vehicle while maintaining the rotational inertia of the engine. For example, in the pull-assist starting mode, when the rope of the recoil starter is pulled by the user, starter-generator controller **110** drives current into stator **118** to apply cranking assist torque to the crankshaft. This will reduce the pull effort required to start the engine.

(32) Starter-generator controller **110** is further configured to provide the available starting mode option to the user via a start enable indicator **116**. Exemplary indicators include audio indicators, visual indicators, tactile indicators, and combinations thereof. For example, the indication may be displayed on a display screen of the vehicle. The indication may be a text notification and/or a light indicator with a blinking pattern and/or color that corresponds to a different starting mode.

(33) In some embodiments, starter-generator controller **110** may receive an indication from the user via user input sensor **114** indicating that the user wants to start the engine. The user input sensor **114** may include, but not limited to, a button, a pull recoil handle sensor, a pull force sensor, and/or a recoil rotation sensor. It should be appreciated that the indication of the user may be received prior to or subsequent to determining the available starting mode of the engine based on the power source level.

(34) Referring now to FIG. 3, a computer-implemented method **200** for controlling different starting modes of an internal combustion engine of a vehicle is shown. As described above, the starting modes of the internal combustion engine includes a full starting mode, a pull-assist starting mode, and a manual starting mode. In the illustrative embodiment, method **200** is performed by a starter-generator controller (e.g., **110**) of a starter-generator system (e.g., **100**) of the vehicle. In block **202**, starter-generator controller **110** receives a user request to start the internal combustion engine. For example, the user request may be received via a user input sensor (e.g., **114**) of the vehicle. User input sensor **114** may include, but not limited to, a button, a pull recoil handle sensor, a pull force sensor, and/or a recoil rotation sensor.

(35) If starter-generator controller **110** determines that a user request has not been received in block **204**, method **200** loops back to block **204** to continue awaiting a user request. If, however, starter-generator controller **110** determines that a user request was received in block **204**, method **200** proceeds to block **206**.

(36) In block **206**, starter-generator controller **110** determines whether a power source level satisfies a first characteristic. For example, the first characteristic may include the power source level being above a first predetermined threshold. It should be appreciated that the power source level (e.g., energy storage capability) may be determined based on various sensor inputs (e.g., battery state-of-charge and battery temperature). The power source level may be determined by the power source **112** or the starter-generator controller **110** based on the embodiment of the power source **112** (e.g., an energy storage device).

(37) If starter-generator controller **110** determines that the power source level satisfies the first characteristic (e.g., above the first predetermined threshold) in block **210**, method **200** proceeds to block **212**. In block **212**, starter-generator controller **110** determines that the starter-generator system **100** supports electric start or restart of the engine by using the power source **112** to deliver electrical power to the ECM **120**. Subsequently, method **200** skips ahead to block **222** to transmit a

notification to the user indicating that the full-starting mode is available to start or restart the vehicle.

(38) Referring back to block **210**, if starter-generator controller **110** determines that the power source level does not satisfy the first characteristic (e.g., below the first predetermined threshold), method **200** advances to block **214**. In block **214**, starter-generator controller **110** determines whether the power source level satisfies a second characteristic. For example, the second characteristic may include the power source level being above a second predetermined threshold. If starter-generator controller **110** determines that the power level satisfies the second characteristic in block **216** (e.g., between the first and second predetermined thresholds), method **200** proceeds to block **218**. In block **218**, starter-generator controller **110** determines that the power is depleted but the remaining power is enough to support the pull-assist starting mode. Subsequently, method **200** skips ahead to block **222** to transmit a notification to the user indicating that the manual starting mode is available to start or restart the vehicle.

(39) Referring back to block **216**, if starter-generator controller **110** determines that the power source level does not satisfy the second characteristic (e.g., below the second predetermined threshold), method **200** proceeds to block **220**. In block **220**, starter-generator controller **110** determines that the power source level is too low and that only the manual starting mode is supported by the starter-generator system **100**. Subsequently, method **200** skips ahead to block **222** to transmit a notification to the user indicating that the pull-assist starting mode is available to start or restart the vehicle.

(40) It should be appreciated that, in some embodiments, starter-generator controller **110** may determine the available starting mode of the engine prior to receiving a user request to start the engine. In other words, blocks **202** and **204** may be performed subsequent to blocks **206-220**.

(41) Referring to FIG. **4**, a hybrid battery system **300** for starting and restarting an engine of a vehicle is described. Hybrid battery system **300** is configured to provide a low self-discharge rate, a high performance at low temperature, a high energy density, and low weight and volume. To do so, hybrid battery system **300** includes a small lithium-ion (Li-Ion) battery **310**, a DC/DC converter **320**, and a supercapacitor system **330**.

(42) Li-ion battery **310** generally has a high energy density and a low self-discharge rate. This allows Li-ion battery **310** to provide electrical power to the engine during several engine starts. However, Li-ion battery **310** generally has a low performance at temperature below 0° C./32° F. Therefore, Li-ion battery **310** is not an efficient energy storage at lower temperature and, thus, is not a reliable energy source to start or restart the engine at lower temperature. Whereas, supercapacitor system **330** is capable of delivering high power even at very low temperatures (e.g., down to -40° C./-40° F.).

(43) In the illustrative embodiment, DC/DC converter **320** and supercapacitor system **330** are part of the vehicle. Small Li-Ion battery **310** may be removably connected to the vehicle. Small Li-Ion battery **310** may be embodied as a 12-voltage energy storage, and DC/DC converter **320** may be embodied as a bidirectional buck-boost converter.

(44) When the Li-Ion battery **310** is connected to the vehicle, it is configured to start pre-charging supercapacitor system **330** for an engine cold start. For example, the pre-charge time may be 10 to 20 seconds. However, it should be appreciated that the pre-charge time may depend on the design of the Li-Ion battery **310** and the DC/DC converter **320**.

(45) Once supercapacitor system **330** is charged, the engine can be started over an inverter and a starter-generator (e.g, alternating current generator (ACG)). Once the engine is running, the starter-generator may be used to recharge supercapacitor system **330** and Li-ion battery **310**. It should be appreciated that, for an engine restart, a pre-charge cycle is not needed and can be done immediately.

(46) Referring to FIG. **5**, an electric reverse control system **400** for restarting an engine in a reverse direction is described. Specifically, upon receiving a reverse control event request from a rider,

electric reverse control system **400** may decelerate a crankshaft of a vehicle and accelerate the crankshaft in the opposite direction to start the engine. To do so, electric reverse control system **400** includes a reverse control user input **410**, an engine control module (ECM) **420**, a starter-generator controller **430**, an integrated starter generator (ISG) **440**, a crankshaft position sensor **450**, a crankshaft **260**, a motor energy storage **470**, an energy storage **480**, and a fuel pump **490**. It should be appreciated that the various controllers described herein may be combined to include the different control sequences described herein and interactions with other components.

(47) Starter-generator controller **430** is operatively coupled to ECM **420** and ISG **440** to control the direction of the engine. ECM **420** is configured to receive a user request from a rider for restarting the engine in a reverse direction (e.g., reversing a direction of an engine rotation) via user input **410**. User input **410** may be a pushbutton or selection on a touchscreen display. When the user request is received, ECM **420** is further configured to determine whether the vehicle is in a predefined condition to restart the engine in the reverse condition. If the user request cannot be completed, ECM **420** may transmit an error message notifying the rider that the vehicle cannot reverse the direction of the engine.

(48) ECM **420** is further configured to disable spark and fuel supplied to the engine in response to a determination that all parameters of the vehicle meet the predefined condition to restart the engine in the reverse condition. Additionally, ECM **420** is configured to transmit a signal to starter-generator controller **430** to switch starter-generator controller **430** to a starter mode. For example, the signal may be sent via a controller area network (CAN), a discrete network, or any other suitable communication channel that may be formed between ECM **420** and starter-generator controller **430**.

(49) Starter-generator controller **430** is configured to switch between the starter mode and a voltage regulator mode to control ISG **440**, which is configured to control crankshaft speed during the reverse control event. For example, in the starter mode, starter-generator controller **430** is configured to decelerate crankshaft **460** in response to receiving an engine reverse request signal from ECM **420**. To do so, starter-generator controller **430** is configured to send excess power to ultracapacitors **470** to apply braking torque to crankshaft **460**. If ultracapacitors **470** is fully charged, starter-generator controller **430** is configured to shunt a stator **118** of ISG **440** of the engine to provide the braking torque to decelerate crankshaft **460**.

(50) When the crankshaft speed drops below a predefined threshold, starter-generator controller **430** is configured to use energy from ultracapacitors **470** to develop torque in the opposite direction of the current crankshaft rotation. This torque will further decelerate crankshaft **460** and spin crankshaft **460** in the opposite direction. It should be appreciated that during the period where starter-generator controller **430** is using energy from ultracapacitors **470**, alternative energy storage **480** (e.g., 12V battery) supplies power to ECM **420** and fuel pump **490**. Once the engine is running in the opposite direction, starter-generator controller **430** is configured to switch back to the voltage regulator mode and recharge ultracapacitors **470**.

(51) Because electric reverse control system **400** allows ISG **440** to restart the engine in either direction, determining correct fuel amount and spark timing may not be necessary to change the engine direction. Accordingly, electric reverse control system **400** improves reliability and performance of a process of reversing the engine rotation.

(52) Referring now to FIGS. **6** and **7**, a computer-implemented method **500** for restarting an engine in a reverse direction by controlling a direction of an engine rotation of a vehicle is shown. In the illustrative embodiment, method **500** is performed in part by an engine control module (ECM) (e.g., **420**) and in part by a starter-generator controller (e.g., **430**) of an electric reverse control system (e.g., **400**) of the vehicle. In block **502**, ECM **420** receives a user request from a rider for reversing a direction of an engine rotation while engine is running. For example, the user request may be received via a user input (e.g., **410**) of the vehicle. User input **410** may be a pushbutton or selection on a touchscreen display. If a user request has not been received in block **504**, method **500**

loops back to block **502** to continue awaiting a user request for reversing the direction of the engine rotation. If, however, a user request has been received, method **500** proceeds to block **506**.

(53) In block **506**, ECM **420** determines if a configuration is satisfied to be reversing the direction of the engine rotation. For example, ECM **420** determines if all parameters that are required to begin reversing the direction of the engine rotation are satisfied. For example, such parameters may include throttle position sensor reading being below a predetermined threshold and/or an engine speed (RPM) being below a predetermined threshold. If ECM **420** determines that the configuration is not satisfied in block **508**, method **500** loops back to block **506** to continue awaiting the configuration to satisfy the predefined conditions. It should be appreciated that, in some embodiments, ECM **420** may notify the rider that the user request cannot be completed. If, however, ECM **420** determines that the configuration is satisfied in block **508**, method **500** proceeds to block **510**.

(54) In block **510**, ECM **420** begins an engine reverse process by disabling spark and fuel to the engine. ECM **420** transmits a signal to starter-generator controller **430** to switch starter-generator controller **430** to a starter mode. For example, the signal may be sent via a controller area network (CAN), a discrete network, or any other suitable communication channel that may be formed between ECM **420** and starter-generator controller **430**.

(55) In response, in block **512**, starter-generator controller **430** begins to decelerate a crankshaft (e.g., **460**) of the vehicle by sending excess power to ultracapacitors (e.g., **470**). This will apply the braking torque to crankshaft **460**. If starter-generator controller **430** determines that ultracapacitors **470** is fully charged in block **514**, method **500** proceeds to block **516**, in which starter-generator controller **430** shunts a stator of ISG **440** of the engine to provide the braking torque to decelerate crankshaft **460** in block **516**. If, however, ultracapacitors **470** is not fully charged, starter-generator controller **430** continues to send excess power to ultracapacitors **470**. Subsequently, method **500** proceeds to block **518** in FIG. 7 as shown by the alphanumeric character A in FIGS. 6 and 7.

(56) In block **518**, starter-generator controller **430** determines whether a crankshaft speed satisfies a predefined characteristic. For example, starter-generator controller **430** determines whether a crankshaft speed is below a predefined threshold. If the crankshaft speed does not satisfy the predefined characteristic (e.g., not below the predefined threshold), method **500** loops back to block **518** to continue determining the crankshaft speed until the crankshaft speed falls below the predefined threshold. If the crankshaft speed satisfies the predefined characteristic (e.g., below the predefined threshold), method **500** proceeds to block **522**.

(57) In block **522**, starter-generator controller **430** uses energy from ultracapacitors **470** to develop torque in the opposite direction of the current crankshaft rotation. This torque will further decelerate crankshaft **460** and spin crankshaft **460** in the opposite direction. During the period where starter-generator controller **430** is using energy from ultracapacitors **470**, alternative energy storage **480** (e.g., 12V battery) supplies power to ECM **420** and fuel pump **490**.

(58) When crankshaft **160** is spinning in the opposite direction, in block **524**, ECM **420** enables spark and fuel to restart the engine. After the engine is running, starter-generator controller **430** switches back to a voltage regulator mode and recharges ultracapacitors **470**, as indicated in block **526**. In the illustrative embodiment, starter-generator controller **430** switches from the starter mode to the voltage regulator mode when certain characteristics (e.g., saturation of motor controller for a predetermined time period, and/or a number of crankshaft revolutions) are satisfied.

(59) Referring to FIG. 8, an engine starting system **700** starting and restarting an engine of a vehicle is described. Engine starting system **700** is configured to provide a high peak power, a high performance at low temperature, a high energy density, a low self-discharge rate, and low weight and volume. To do so, engine starting system **700** includes an integrated starter generator (ISG) **710**, a bi-directional inverter **720**, and lithium-ion capacitors **730**.

(60) Lithium-ion capacitors **730** is configured to provide power to start and restart an engine of a vehicle. Once the engine is running, integrated starter generator **710** may be used to recharge

lithium-ion capacitors **730**. When lithium-ion capacitors **730** is charged, the engine can be started over bi-directional inverter **720** and integrated starter generator **710**.

(61) Lithium-ion capacitors **730** may be rapidly charged and discharged, which provides a high peak power. Lithium-ion capacitors **730** has enough energy to provide power for several engine starts. Additionally, lithium-ion capacitors **730** has a low self-discharge rate and keeps the energy. In other words, lithium-ion capacitors **730** does not need to be pre-charged and may be used for an engine cold start.

(62) Engine starting system **700** offers durability because lithium-ion capacitors **730** are not flammable and maintenance free, thereby providing a safe storage for engine starting system **700**. By replacing a lead-acid battery with lithium-ion capacitors **730**, the overall weight of the vehicle is significantly reduced compared to a starter system with lead-acid battery.

(63) Referring to FIG. **9**, a starter-generator system **800** for providing electric start, electrically assisted-starting, and power generation is described. Starter-generator system **800** includes a current steering integrated starter-generator (ISG) controller **804**, an integrated starter-generator (ISG) **802**, an ultracapacitor module (UCM) **810**, and an energy storage **822**. Ultracapacitor module **810** further includes ultracapacitors **812** and a bidirectional DC/DC voltage converter **814**. In the illustrative embodiment, bidirectional DC/DC voltage converter **814** is combined with ultracapacitors **812** into a single ultracapacitor module **810**. However, in some embodiments, bidirectional DC/DC voltage converter **814** may be combined with ISG controller **804** or as a standalone module.

(64) Ultracapacitors **812** are configured to store energy and supply high power during engine starting. Whereas, energy storage **822** of the vehicle is configured to store electrical energy when the vehicle is not in use (e.g., engine stopped). For example, energy storage **822** may be embodied as a lightweight removable battery (e.g., 12V battery). Additional details of features of removable battery may be disclosed in U.S. Pat. No. 10,731,620, issued Aug. 4, 2020, and entitled "BATTERY KEY, STARTER AND IMPROVED CRANK", the complete disclosure of which is expressly incorporated by reference herein. It should be appreciated that, in some embodiments, starter-generator system **800** may not include a 12V energy storage. In such embodiments, the engine may be started manually if there is insufficient charge stored in ultracapacitors **812**.

(65) Integrated starter generator (ISG) **802** mechanically coupled to an engine crankshaft and is configured to drive current into a stator of ISG **802** to apply cranking torque to the crankshaft to start the engine. In some embodiments, starter-generator system **800** may include a crankshaft position sensor (CPS) **816** to measure a flywheel electrical angle.

(66) ISG controller **804** is configured to operate as a motor controller **806** during engine starting and as a current steering **808** afterwards. In the illustrative embodiment, current steering **808** is embodied as a voltage regulator with current-steering functionality. Current steering **808** is configured to direct rectified ISG current to one of three outputs at a time in order to maintain all three at their target levels. For example, the three outputs may include the 16V ECM power output, 14V critical output, and 29V DC link output. In other words, ISG controller **804** is configured to apply power to the stator for starting and to take power from the stator when running to supply the vehicle's electrical loads.

(67) Bidirectional DC/DC voltage converter **814** is configured to provide efficient charging of ultracapacitors **812** from zero voltage to a voltage greater than a voltage of battery **822**. To do so, bidirectional DC/DC voltage converter **814** is configured to charge ultracapacitors **812** prior to engine starting, supply power directly from energy storage **822** (e.g., 12V battery) to the ISG controller **804** during the engine starting, and recharge an energy storage **822** and supply vehicle electrical loads once the engine has started. By charging ultracapacitors **812** in this manner, the peak current required from the vehicle battery **822** may be reduced and the cost may also be reduced compared to a conventional electric start and magneto power generation system.

(68) In some embodiments, charging of ultracapacitor **812** may begin in response to receiving a

user input. For example, the user may press a button to initiate charging of ultracapacitor **812**. Upon receiving the user input, ISG controller **804** may determine an ultracapacitor level and start the engine as soon as it determines that sufficient energy is stored in ultracapacitor **812** to crank the engine. It should be appreciated that, in some embodiments, the ultracapacitor level may be continually or periodically determined prior to receiving the user input.

(69) In some embodiments, charging of ultracapacitor **812** may begin as soon as the vehicle is turned on via a key switch **828** or other means. In such embodiments, the vehicle may indicate when starter-generator system **800** is ready to start the engine (e.g., notifies the user of an available start mode). In response, the user may press a button or rotates the key to a start position to initiate engine start.

(70) In some embodiments, ISG controller **804** uses multiple sensors in combination to determine when starter-generator system **800** is ready to start the engine. For example, ISG controller **804** may use measurements of engine and/or ambient temperature, ultracapacitor bank voltage, and/or DC/DC converter current. ISG controller **804** may also consider the time history of these values, for example, noting when the ultracapacitor bank voltage exceeds a predetermined threshold and then waiting a predetermined period of time based on the engine temperature. It should be appreciated that ISG controller **804** may measure these values directly or receive them from UCM **810** or ECM **824** via digital communication means.

(71) In some embodiments, ISG controller **804** may produce torque to assist manual pull starting of the engine. In such embodiments, a pushbutton may be integrated into a rope handle such that when the operator depresses the button and pulls on the rope, ISG controller **804** sends current to ISG **802** to produce torque in the forward direction. Alternatively, starter-generator system **800** may include a sensor for detecting when there is tension on the starting rope and initiating starting in response.

(72) It should be appreciated that, in some embodiments, starter-generator system **800** may also be used to facilitate reversing the direction of engine rotation when used on 2-stroke engines, as described above in FIGS. 5-7.

(73) Referring now to FIG. **10**, an exemplary schematic circuit diagram of ISG controller **804** is shown. ISG controller **804** is configured to convert electrical power between a stator of ISG **802** and three-DC voltage buses. For example, inputs of ISG controller **804** include Switched Power, ECM Remote Sense, and Encoder inputs A-C. Switched Power is connected to a battery through a key switch (e.g., **828**) to power-on ISG controller **804**. ECM Remote Sense is an optional analog input to measure the ECM output at a point outside of ISG controller **804** in order to improve voltage control accuracy. Encoder inputs A-C are generated by external sensors based on a rotor position.

(74) Outputs of ISG controller **804** include ECM **824** and Encoder Output. ISG controller **804** is configured to supply voltage to ECM **824**. Encoder Output is an optional signal that may change state whenever one of the encoder inputs change state, for use by ECM **824** as a high resolution engine speed signal.

(75) Additionally, bidirectional signals may be generated. For example, stator voltages are wired to a three-phase stator that may be Wye/Star or Delta-wound. VCAP is a power source input during engine start and a power output while the engine is running. CANH or CANL is a standard communications bus for transmitting status information, diagnostics, and reprogramming.

(76) As shown in FIG. **10**, ISG controller **804** has two main sections (i.e., a rectify/boost/start (RBS) section and a current steering section). Rectify/Boost/Start (RBS) section is responsible for converting between 3 phase AC and DC power. Current Steering section is responsible for directing power from a Vcap pin to RBS section during engine starting and for directing power from that section to ECM, Critical, and Vcap pins while the engine is running. In addition, control circuitry is configured to operate both sections based on the various inputs.

(77) In RBS section, MOSFETs Q1-Q6 form a conventional three-phase bridge. Optional resistors

R1 and R2 allow for stator current sensing when Q1 or Q3 respectively are on. Stator current sensing may also be accomplished with resistors in series with the stator leads or with Hall-effect or other current sensors on the stator leads. MOSFETs are controlled via signals from the control circuitry.

(78) In Current Steering section, MOSFETs Q7 and Q8 are individually controlled. When Q7 is on current is allowed to flow from the RBS section to the Vcap pin through the body diode of Q8. When Q8 is on, current is allowed to flow in the opposite direction. MOSFETS Q9 and Q10 are controlled as a pair. When on, current flows from the RBS section to the Critical output. MOSFET Q11 is turned on to allow current to flow from the RBS section to the ECM pin through diode D2. Diodes D3 and D4 supply power from the Switched Power input to the ECM and Critical outputs prior to engine start. Once the engine has started, the ECM and Critical voltages will rise, blocking current from flowing through these diodes. Resistor R3 allows the GND pin current to be measured, and Capacitors C2-C4 provide filtering.

(79) ISG control circuit is configured to measure voltage at the ECM, Critical, and Vcap pins as inputs to its control algorithm. Additionally, ISG control circuit may also measure the stator voltages to determine flywheel RPM or use the Encoder inputs. ISG control circuit includes 5 states: Wait, Engine Start, Engine Reverse Start, Boost, and Power Generation. In Wait state, all MOSFETs are off. In Engine Start state, RBS Section operates to produce starting torque and Current Steering Section operates supply power from Vcap. In Engine Reverse Start state, RBS Section operates to produce starting torque in the reverse direction, and Current Steering Section operates to supply power from Vcap. In Boost state, Current Steering Section operates to supply electrical power to the ECM, Critical, and Vcap pins during a manual start. For example, ISG control circuit is in Boost state during a pull-assist starting mode, which supports electric start with manual pull assist. Lastly, in Power Generation state, Current Steering Section operates to supply electrical power to the ECM, Critical, and Vcap pins.

(80) ISG controller 804 keeps all MOSFETs off during the Wait state. Subsequently, ISG controller 804 transitions to the Boost state when the Encoder inputs indicate that the flywheel is rotating or AC voltage is detected on the stator leads. Alternatively, ISG controller 804 may transition to the Boost state in response to a signal received over the CAN bus.

(81) Engine Start is accomplished by turning on Q8 and optionally Q7 (to reduce the voltage drop across Q7), and controlling the RBS section as a typical BLDC motor controller, with commutation control of Q1-Q6 and PWM control of Q4-Q6 to achieve a target motor current based on flywheel angle determined by the Encoder inputs, as shown in Table 1.

(82) TABLE-US-00001 TABLE 1 Engine Start Operation Encoder Q7- Q9- ABC Q1 Q2 Q3 Q4 Q5 Q6 Q8 11 010 ON OFF OFF OFF PWM PWM ON OFF 110 ON ON OFF OFF OFF PWM ON OFF 100 OFF ON OFF PWM OFF PWM ON OFF 101 OFF ON ON PWM OFF OFF ON OFF 001 OFF OFF ON PWM PWM OFF ON OFF 011 ON OFF ON OFF PWM OFF ON OFF

(83) ISG controller 804 transitions to the Power Generation state after a predetermined time has passed since the beginning of engine start. Alternately, ISG controller 804 may transition to the Power Generation state when an error between the target motor current and the actual motor current exceeds a predetermined threshold for a predetermined period. ISG controller 804 transitions to the Wait state when the engine RPM falls below a predetermined threshold.

(84) Moreover, Engine Reverse Start is accomplished by turning on Q8 and optionally Q7 (to reduce the voltage drop across Q7), and controlling the RBS section as a typical BLDC motor controller, with commutation control of Q1-Q6 and PWM control of Q4-Q6 to achieve a target motor current based on flywheel angle determined by the Encoder inputs, as shown in Table 1b.

(85) TABLE-US-00002 TABLE 1b Engine Reverse Start Operation Encoder Q7- Q9- ABC Q1 Q2 Q3 Q4 Q5 Q6 Q8 11 010 OFF ON ON PWM OFF OFF ON OFF 110 OFF OFF ON PWM PWM OFF ON OFF 100 ON OFF ON OFF PWM OFF ON OFF 101 ON OFF OFF OFF PWM PWM ON OFF 001 ON ON OFF OFF OFF PWM ON OFF 011 OFF ON OFF PWM OFF PWM ON

OFF

(86) ISG controller **804** transitions to the Power Generation state after a predetermined time has passed since the beginning of Engine Reverse Start. Alternately, ISG controller **804** may transition to the Power Generation state when the error between the target motor current and the actual motor current exceeds a predetermined threshold for a predetermined period. ISG controller **804** transitions to the Wait state when the engine RPM falls below a predetermined threshold.

(87) Additional details of features of a power boosting regulator may be disclosed in U.S. Pat. No. 10,793,226, issued Oct. 6, 2020, and entitled "POWER BOOST REGULATOR", U.S. Pat. No. 10,870,465, issued Dec. 22, 2020, and entitled "POWER BOOST REGULATOR", U.S. Pat. No. 10,780,949, issued Sep. 22, 2020, and entitled "POWER BOOST REGULATOR", and U.S. Pat. No. 10,974,790, filed Apr. 13, 2021, and entitled "POWER BOOST REGULATOR", the complete disclosures of which are expressly incorporated by reference herein. Operation in the Power Generation state, control period Tctrl is divided into 4 parts: Tecm, Tcritical, Tcap, and Tshunt. Each of the first three time periods is increased when the respective pin is below the target voltage for that pin and decreased when the voltage is high. Tshunt is set to the amount of time remaining in Tctrl. A non-negative minimum value for Tshunt (Tmin) is predetermined based on engine RPM. The outputs are prioritized as follows:

(88) 1. If $(T_{ecm} + T_{min}) > T_{ctrl}$, set $T_{shunt} = T_{min}$, $T_{ecm} = T_{ctrl} - T_{min}$, and set Tcritical and Tcap to zero

(89) 2. Else if $(T_{ecm} + T_{min} + T_{critical}) > T_{ctrl}$, set $T_{shunt} = T_{min}$, set $T_{critical} = T_{ctrl} - T_{min} - T_{ecm}$, and set Tcap to zero

(90) 3. Else if $(T_{ecm} + T_{min} + T_{critical} + T_{cap}) > T_{ctrl}$, set $T_{shunt} = T_{min}$ and set $T_{cap} = T_{ctrl} - T_{min} - T_{ecm} - T_{critical}$

(91) 4. Else set $T_{shunt} = T_{ctrl} - T_{ecm} - T_{critical} - T_{cap}$

(92) In each time period ISG controller **804** is operated in accordance with Table 2.

(93) TABLE-US-00003 TABLE 2 Power Generation Operation Time Period Q1-Q3 Q7 Q8 Q9-10 Q11 Tecm OFF ON OFF OFF ON Tcritical OFF ON OFF ON OFF Tcap OFF ON ON (opt.) OFF OFF Tshunt ON ON OFF OFF OFF

(94) ISG controller **804** transitions to the Boost state if Vcap falls below the ECM target voltage. ISG controller **804** transitions to the Wait state if the engine RPM falls below a predetermined threshold or a Direction Reversal command is received over CAN. In an alternative implementation, MOSFETs Q1-Q6 are controlled based on the Encoder inputs and Table 1 when not in the Tshunt time period.

(95) In the Boost state ISG controller **804** is operated as defined by Table 3, following the same prioritization rules for the time periods defined for the Power Generation state. Control in this manner maintains the prioritization of the ECM and Critical outputs over Vcap. In an alternative implementation, MOSFETs Q1-Q6 are controlled based on the Encoder inputs and Table 1 or 1b when not in the Tshunt time period.

(96) TABLE-US-00004 TABLE 3 Boost Operation Time Period Q1-Q3 Q4-6 Q7 Q8 Q9-10 Q11 Tecm OFF OFF OFF OFF OFF ON Tcritical OFF OFF OFF OFF ON ON Tcap OFF OFF ON ON (opt.) OFF ON Tshunt ON OFF OFF OFF OFF ON

(97) ISG controller **804** transitions to the Power Generation state when Vcap exceeds the ECM target voltage. ISG controller **804** transitions to the Wait state when the engine RPM falls below a predetermined threshold. Optionally, ISG controller **804** transitions to the Engine Start state when the engine controller sends a direction reversal command.

(98) Referring now to FIG. 11A, an exemplary schematic circuit diagram of ultracapacitor module **810** is shown. In the illustrative embodiment, ultracapacitor Bank **812** and DC-DC Converter **814** are combined into one module **810**. Ultracapacitor Module (UCM) **810** includes a DC/DC converter **814** to charge ultracapacitors **812** at 62 A max input current and then supply up to 41 A at 14.5V to charge the battery **822** and power chassis loads **818**. DC/DC converter **814** continues to

supply current during engine starting, thereby reducing the size of the capacitor bank.

(99) Ultracapacitor Bank **812** consists of multiple ultracapacitors wired in series (C1-C12) to increase the operating voltage. Ultracapacitors may also be wired in parallel to increase energy storage and increase power delivery. Resistors R1-R12 are present in Ultracapacitor Bank **812** to balance the voltage across the ultracapacitors to prevent overcharging of one capacitor in the string. Resistors are selected to dissipate between 0.01 and 1.0 W from a fully charged ultracapacitor. DC/DC converter **814** is composed of a half-bridge arrangement of MOSFETs Q1 and Q2, inductor L1, MOSFET Q5, diode D1, filter capacitors C14 and C15, and Control Logic.

(100) In the illustrative embodiment, DC/DC converter **814** has four Modes: Start mode, Mode 1a, Mode 1b, and Mode 2. In the Start mode, DC/DC converter **814** waits for a request signal to begin operation in either Mode 1a or 1b. However, the request signal may be an optional features. In Mode 1a, voltage VCAP is less than VBAT and DC/DC converter **814** steps down the VBAT voltage to charge the Capacitor Bank. In Mode 1b, VCAP is at least VBAT and DC/DC converter **814** steps up the VBAT voltage to charge the Capacitor Bank **812**. In Mode 2, VCAP is greater than VBAT and DC/DC converter **814** steps down the VCAP voltage to supply power to VBAT.

(101) At start-up, ultracapacitors may be fully discharged. In such case, UCM **810** begins operation by PWM control of MOSFET Q5. This is Mode 1a. When Q5 is on, current flows from battery **822** through Q5, L1, Q1, and Ultracapacitor Bank **812**. Inductor L1 limits the current rise. When Q5 is shut off, current flows through D1 instead of Q5 and L1 limits the current decay.

(102) Once the voltage across the Ultracapacitor Bank (VCAP) has risen to nearly the level of the battery (VBAT), DC/DC converter **814** switches to Mode 1b. In Mode 1b, Q5 is turned fully-on (100% duty), and Q1 and Q2 are PWMed in a complimentary fashion such that only one is on at a time. When Q2 is on L1 is effectively connected across battery **822** and current rises. When Q1 is on, L1 supplies current to the ultracapacitors **812** and L1's current decays. In this way, the Ultracapacitor Bank is charged to a voltage greater than the Battery Voltage.

(103) In Mode 2, UCM **810** controls MOSFET Q5 on 100% of the time and PWMs Q1 and Q2 with complementary signals. The VCAP bus is at a higher voltage than the VBAT bus, so when Q1 is on current flows from VCAP through Q1, L1, and Q5 to VBAT, and the current in L1 climbs. When Q2 is on current flows from ground through Q2, L1, and Q5 and the current in L1 falls.

(104) In the illustrative embodiment, ISG Controller **804** regulates the DC link voltage VCAP to 29V when the engine is running and Converter **814** operates in Mode 2 with Q1 and Q2 PWMed in complementary fashion at a fixed 50% duty cycle. In this configuration, the VBAT output of the Converter equals the VCAP input times the duty cycle, such that 14.5V is supplied to Battery **822**. The fixed duty cycle prevents interactions with VCAP voltage regulation and allows ISG Controller **804** to reduce the power supplied to Battery **822** when there is insufficient total power available from ISG **802** to meet the total required by the three ISG Controller outputs.

(105) UCM **810** may disable PWM control or reduce the duty cycle of Q1 if the VCAP voltage falls below a predetermined voltage, typically 17 V. This maintains VCAP voltage above VECM and keeps ISG Controller **804** in the normal operating mode.

(106) In one embodiment, a Control Logic in DC/DC converter **814** monitors voltage VECM to determine the mode to operate in. When VECM is between 6 and 15V, DC/DC converter **814** charges Ultracapacitor Bank **812** in either Modes 1a or 1b, based on voltage VCAP relative to voltage VBAT. When the Control Logic measures greater than 15V on VECM it transitions to Mode 3. In another embodiment, a state of a digital or serial signal from ISG controller **804** indicates whether DC/DC converter **814** should operate in Mode 1 or Mode 2.

(107) The System may be used without battery **822** if the vehicle has a manual starting option, such as a rope-pull starter. In this case, the voltage on VECM transitions from near zero volts to greater than 15V and DC/DC converter **814** transitions directly to Mode 2.

(108) FIG. **11B** illustrates key voltages and currents in UCM Modes 1 and 2. In Mode 1a, the Ultracapacitor Bank voltage Vcap (in red) is less than Battery voltage Vbat (in green), and the DC-

DC converter current (dark blue) is at its maximum. This current charges the Ultracapacitor Bank **812**. In Mode 1a, the Battery current I_{bat} (light blue) rises towards its maximum. At the transition between Modes 1a and 1b, the two voltages are equal as are the two currents. In Mode 1b, DC/DC converter **814** operates with fixed battery current, which results in nearly constant input power. The output current falls as V_{cap} increases due to the limited input power. At a point determined by ISG controller **804**, it begins to generate torque to turn the crankshaft, drawing power from DC/DC converter **814** and Ultracapacitor Bank **812**. DC/DC converter **814** continues to operate in Mode 1b during and following engine start in order to maximize the power available for starting and to prepare for a second engine start event if needed. When the predetermined condition is reached (VECM above a threshold voltage or upon a signal from ISG Controller **804**), UCM **810** transitions to Mode 2.

(109) As Mode 2 operation has been described, the current input into the VCAP terminal of UCM **810** is zero when **Q1** is off and equal to the inductor current when **Q1** is on. Since ISG Controller **804** cannot react quickly enough to switch its output current on and off during each cycle, large capacitors in ISG Controller **804** and in DC/DC Converter **814** are required to provide filtering. This is the function of **C15** in the schematic shown in FIG. **11A**.

(110) In the illustrative embodiment, DC/DC Converter **814** is a multiphase converter with two phases. In this configuration, there are two inductors **L1** and **L2** and two half bridges (**Q1+Q2** and **Q3+Q4**). Initial charging of Ultracapacitor Bank **812** is the same as the charging described above, but the current is carried by both inductors and both **Q1** and **Q3**.

(111) When **Q5** reaches 100% duty cycle, the converter logic begins PWM of both half bridges at the same frequency but with a 180° offset. For example, if the PWM period is 2 μ s (500 kHz switching frequency), then the turn-on of **Q1** and **Q3** is offset by 1 μ s. **Q2** is controlled complementary to **Q1** and **Q4** is controlled complementary to **Q3**. This is configured to reduce voltage and current ripple, as it effectively doubles the PWM switching frequency. This reduction reduces the minimum size of capacitors **C14** and **C15**.

(112) When the engine is running and ISG Controller **804** is regulating the DC link voltage VCAP to 29V, **Q1** and **Q3** are controlled at 50% duty cycle, such that each time **Q1** is turned off **Q3** is turned on, and vice-versa. **Q2** and **Q4** continue to be controlled complementary to their respective counterparts. In this way, current from VCAP is alternately fed to **L1** and **L2** without interruption, further reducing voltage and current ripple. At all times the current into DC/DC Converter **814** is near its average value, varying only with the rise and fall of the inductor currents.

(113) Referring to FIGS. **12A** and **12B**, a removable lithium-ion battery **900** is described. Lithium-ion battery **900** is removably coupled to a vehicle to perform various functions. For example, removable lithium-ion battery **900** may be used off-board charging, higher voltage for integrated-starter-generator systems, a bidirectional DC/DC converter for charging off of vehicle power when the engine is running, and/or powering vehicle electrical systems at other times. Additionally, removable lithium-ion battery **900** may include a pushbutton and a plurality of LEDs for a status and state-of-charge indication. For example, in some embodiments, a single press of the pushbutton may cause the removable battery **900** to display the state-of-charge by turning on one or more of the plurality of LEDs for a period of time (e.g., 5 seconds). Additionally, a double press of the pushbutton may cause the removable battery **900** to turn on or off a white LED for use as a flashlight. A triple press of the pushbutton may cause the battery to turn on or off a self-heating mode to function as a hand warmer and/or preheat the removable battery **900** prior to be coupled to the vehicle.

(114) To do so, lithium-ion battery **900** includes a housing **902** that includes a USB port **904** and multiple cells **906** in series configuration with integral cell heating element **908**, charge circuit, and/or a serial communication pin. Lithium-ion battery **900** further includes a bidirectional DC/DC converter (e.g., **1012** in FIG. **13**) to allow lithium-ion battery **900** to charge from vehicle power when the engine is running, supply power to the vehicle for fuel pump and engine controller prior

to starting, and power in-vehicle infotainment system and other connected vehicle functions when the key is off.

(115) Additionally, or alternatively, in some embodiments, lithium-ion battery **900** may further include a button (not shown) and light-emitting diode (LED) lights (not shown) to indicate a battery state of charge. For example, when a user pushes the button, the LED lights illuminate to indicate the battery state of charge. Additionally, or alternatively, when the user taps the button three-times, battery pre-heat or hand warmer function using the integral cell heating element **908** may be activated. Additionally, or alternatively, lithium-ion battery **900** may include optional LED flashlight.

(116) Additionally, or alternatively, lithium-ion battery **900** may be used with an integrated starter-generator (ISG) (e.g., **1008** in FIG. **13**) to provide higher voltage to improve starting torque. Additionally, or alternatively, the serial communication link may be used to allow an ISG controller (e.g., **1010** in FIG. **13**) to control the integral cell heating element **908** and the disconnect switch. In some embodiments, removable lithium-ion battery **900** may further include security functionality via secure communications with a vehicle.

(117) Additionally, or alternatively, lithium-ion battery **900** may include a disconnect switch (not shown) to remove battery voltage from terminals when the battery is not installed. As described further below, lithium-ion battery **900** may include four connections to connect to start power, constant power, serial communication via local interconnect network (LIN), and ground.

(118) Referring now to FIG. **13**, a removable lithium-ion battery system **1000** for supplying power to an integrated starter-generator (ISG) controller for vehicle starting is described. Removable starting battery system **1000** includes an integrated starter-generator (ISG) controller **1010**, an integrated starter-generator (ISG) **1008**, and a bidirectional DC/DC converter module **1012**.

Lithium-ion battery **900** is operatively coupled to integrated starter-generator (ISG) **1008** to provide higher voltage to improve starting torque. ISG controller **1010** is configured to operate as a motor controller **1004** during engine starting and as a current steering **1006** afterwards. In the illustrative embodiment, current steering **808** is embodied as a voltage regulator with current-steering functionality.

(119) Lithium-ion battery **900** may transmit status data to ISG controller **1010** including state-of-charge, temperature, battery voltage, and current of lithium-ion battery **900**. Lithium-ion battery **900** may also calculate and transmit its power capacity based on its temperature and state-of-charge.

(120) Additionally, ISG controller **1010** is operatively coupled to removable lithium-ion battery **900** to control operations of the lithium-ion battery **900**. For example, ISG controller **1010** may be configured to control integral cell heating element **908** and the disconnect switch via a serial communication link. The serial communication link allows for bidirectional communication with ISG controller **1010** using a local interconnect network (LIN) **1024** or a similar protocol.

(121) Additionally, ISG controller **1010** is configured to transmit commands to lithium-ion battery **900** to change a battery operating state of lithium-ion battery **900** to facilitate engine starting. To do so, removable lithium-ion battery **900** supports different battery operating states, as illustrated in FIG. **14**. For example, the battery operating states include an off state **1102**, a wake state **1104**, a pre-charge state **1106**, a power delivery state **1108**, and a run state **1110**. Additionally, as described above, lithium-ion battery **900** includes four terminals or connections: battery or start power (e.g., 29V battery power supplied to a DC link capacitor **1002** of ISG controller **100** and/or a chassis **1022** of a vehicle via DC/DC converter module **1012**), constant power (e.g., 12V), a serial communication (e.g., via a local interconnect network (LIN) **1024**), and ground (not shown).

(122) For example, in off state **1102**, lithium-ion battery **900** is configured to supply power to a vehicle at reduced voltage to reduce battery drain. In wake state **1104**, lithium-ion battery **900** is configured to supply constant power (e.g., 12V power) to the vehicle to power ECM **1018** and a fuel pump (e.g., **1020**). In pre-charge state **1106**, lithium-ion battery **900** is configured to supply

12V power to the vehicle (e.g., chassis **1022**) and supply a capacitor charging current to ISG controller **1010** in preparation for connecting battery voltage to ISG controller **1010**. In power delivery state **1108**, lithium-ion battery **900** is configured to supply both 12V and 29V power to the vehicle for engine starting. In run state **1110**, lithium-ion battery **900** is configured to receive 12V power from the vehicle for charging and to maintain optimum operating temperature. To do so, in the illustrative embodiment, lithium-ion battery **900** is configured to transmit status data to ISG controller **1010** including state-of-charge, temperature, battery voltage, and current of lithium-ion battery **900**. Lithium-ion battery **900** may also be configured to calculate and transmit its power capacity based on its temperature and state-of-charge.

(123) Additionally, the battery operating states include an off-board charging state **1112** and an off-board heating state **1114** for controlling off-board charging feature and off-board heating feature of lithium-ion battery **900**, respectively.

(124) It should be appreciated that removable lithium-ion battery **900** including one or more features described herein is configured to reduce weight compared to a conventional battery, improves cold-start capability, and reduces battery maintenance during off-season.

(125) Referring to FIG. 15, an alternative embodiment of a removable lithium-ion battery system **1200** for supplying power to an integrated starter-generator (ISG) controller for vehicle starting is described. An exemplary schematic circuit diagram of removable lithium-ion battery system **1200** is shown in FIGS. 17A and 17B. Removable lithium-ion battery system **1200** include components similar to those components included and described in removable lithium-ion battery system **1000**. In the illustrative embodiment, a high voltage, high power removeable lithium-ion battery **1230** provides power to an integrated starter-generator (ISG) controller **1210**, which supplies power to a fuel pump **1222**, an engine control module (ECM) **1224**, and an integrated starter-generator (ISG) **1222** to start the engine. As described further below, removeable lithium-ion battery **1230** may be used in order to reduce weight, extend storage time, and prevent over-discharge damage.

Removeable lithium-ion battery **1230** allows removable lithium-ion battery system **1200** to perform instant starting and does not require charging or heating time prior to engine start. Additionally, removable lithium-ion battery system **1200** does not require battery maintenance between riding seasons. Removable lithium-ion battery system **1200** does not need harness diodes, capacitors, or relays and supports telematics.

(126) ISG controller **1210** includes current steering, motor control, and high-side PWM functionality. For example, ISG controller **1210** is configured to operate as a motor controller **1218** during engine starting and as a current steering **1216** afterwards. In the illustrative embodiment, current steering **808** is embodied as a voltage regulator with current-steering functionality. Additionally, ISG may be configured to perform a secure identification of lithium-ion battery **1230** via LIN **1238** when a power button **1234** is pressed.

(127) Removeable lithium-ion battery **1230** is configured to supply power to ISG controller **1210** for vehicle starting. In some embodiments, removeable lithium-ion battery **1230** may function as a key **1236**. Removeable lithium-ion battery **1230** has a voltage greater than a typical vehicle starting battery. Lithium-ion battery **1230** is also configured to supply nominal 12V power to the vehicle when the engine is not running. Removeable lithium-ion battery **1230** uses its state-of-charge and temperature to calculate a maximum discharge current, which is sent to ISG controller **1210**. ISG controller **1210** uses the engine temperature and time stopped to determine required starting power. When the required starting power is less than the available power, lithium-ion battery system **1200** indicates that the engine is ready to start.

(128) In some embodiments, removable lithium-ion battery system **1200** allows for batteryless manual starts when the required starting power is more than the available power and/or when removeable battery **1230** is absent. In some embodiments, a batteryless start or pull start may be initiated by the operating pulling the start rope. This spins the flywheel, which produces voltage and current in the stator and powers ISG controller **1210**. In response, ISG controller **1210** may

determine that a pull-start is in process by measuring a frequency of the AC stator voltage and/or observing a changing position reported by a crankshaft position sensor (CPS) **1220**.

(129) Additionally, ISG controller **1210** may shunt the stator current to build current at a minimum duty cycle determined from the flywheel RPM. When not being shunted, current flows to the current-steering circuitry. As shown in the exemplary schematic circuit diagram in FIGS. **17A** and **17B**, MOSFET **Q3** may be turned on to charge **C3** and power the engine controller. The body diode in **Q2** may cause current to also flow to charge **C2**. Additionally, MOSFET **Q1** may be turned on at a high initial duty cycle ($>95\%$) $DC_{sub.0}$ to supply maximum power to fuel pump **1222** while **C2** is being charged. To do so, ISG controller **1210** may begin PWM operation of MOSFET **Q1** to power fuel pump **1222** from removable battery **1230**, ramping the duty cycle from zero to $\min(DC_{sub.0}, 14V/V_{C2})$, where $DC_{sub.0}$ is the maximum duty cycle for the **Q1** drive circuitry and V_{C2} is the voltage on capacitor **C2**. When the ECM voltage rises to a predetermined threshold, MOSFET **Q3** may be turned off for part of each cycle to direct more current to **C2**. Furthermore, when the voltage on **C2** reaches a predetermined threshold, the shunting duty cycle of the three-phase bridge may be increased to prevent overcharging of **C2** and **C3**. When a predetermined time has passed since the start of the pull start, ISG controller **1210** may transition to the run state.

(130) In the Run state, ISG controller **1210** uses a fixed PWM period T . Each PWM period is divided into a time $T1$ where the stator is shunted by the three-phase bridge, a time $T2$ where current flows through **Q2**, and times $T3$ - $T5$ where current flows through **Q3**-**Q5** respectively. PID control loops determine times $T2$ - $T5$ to maintain each output at its predetermined thresholds. The flywheel RPM determines a minimum shunting time $T1_{min}$. When the sum $(T1_{min} + T2 + \dots + T5)$ is less than T , $T1$ is set to $T - (T2 + \dots + T5)$. When the sum $(T1_{min} + T2 + \dots + T5)$ is greater than T , $T1$ is set to $T1_{min}$ and $T5$ is reduced so that $(T1 + \dots + T5)$ is equal to T . Since no time periods can be negative, if necessary $T5$ is set to 0 and $T4$ is reduced. This continues until $T = (T1 + \dots + T5)$.

(131) When removable battery **1230** is absent, the target voltage on **C2** is set to twice the target fuel pump voltage to produce a 50% duty cycle on **Q1** so that **D1** and **D7** see equal average current. ISG controller **1210** may determine that removable battery **1230** is absent if data is not received from removable battery **1230** via LIN bus **1238**. When removable battery **1230** is present but too cold to accept charge, the target voltage on **C2** is set to a voltage that is lower than the pack voltage to prevent charging of the removable battery, typically the same voltage as for the no-battery condition. When removable battery **1230** is present and able to accept current, the voltage on **C2** is first raised to the pack voltage. ISG controller **1210** then determines the average current draw through **Q1** as the current out of the 3 phase bridge times the duty cycle of **Q2**.

(132) In some embodiments, removable lithium-ion battery system **1200** includes a fixed lithium-ion battery **1232** to provide power when the removable lithium-ion battery **1230** is absent. Fixed lithium-ion battery **1232** may include charge/discharge circuitry, a temperature sensor, a battery cell, and heating circuitry. For example, in the off state, fixed lithium-ion battery **1232** is configured to output 8V with a current limit of approximately 100 mA. When the voltage on its Constant Power pin is greater than 13V and the battery is too cold to charge, it turns on a heater, powered from the Constant Power pin. When an acceptable temperature is reached, it takes power from the Constant Power pin to recharge. An optional LIN serial connection may be used to provide status information to ISG controller **1210**, such as temperature, SOC, heater status, and any detected fault conditions.

(133) An exemplary circuitry of fixed lithium-ion battery **1232** is shown in FIG. **18**. The illustrative design includes a single battery cell for simplicity, which eliminates the need for cell balancing and ensures that the cell voltage is lower than the power pin. However, it should be appreciated that, in some embodiments, the fixed lithium-ion battery **1232** may include multiple battery cells. As shown in FIG. **18**, MOSFETs **Q4** and **Q5** and inductor **L1** form a bi-directional DC-DC converter for charging and discharging of battery cell **B1**. Electrical heater **R1** is included to raise the cell

temperature as needed. The heater is powered from the power pin, such that it is not limited by the discharge current limit of a cold battery cell. Additionally, MOSFET Q2 is normally on but may be turned off to prevent over discharge of the battery cell.

(134) Moreover, when the voltage on the constant power pin is greater than 13V and the battery is too cold for charging, the control logic turns on MOSFET Q1 to apply power to the heater. Once the temperature is in an acceptable range, it varies the duty cycle of MOSFET Q5 to provide a controlled charge current to B1. When the voltage on the constant power pin is less than 8V, the control logic varies the duty cycle of MOSFET Q4 to transfer power from B1 to the constant power pin in order to maintain 8V at the constant power pin. The output current is measured through current sensor R2 and the voltage on the constant power pin is allowed to drop to prevent the output current from exceeded a predetermined threshold. In some embodiments, the control logic may provide status information over a serial bus, including cell voltage, SOC, and temperature, output current, and fault status.

(135) Referring back to FIGS. 17A and 17B, an exemplary schematic circuit diagram of removable lithium-ion battery system **1200** is shown. Removeable battery circuitry includes four electrical connections, including Start Power, Constant Power, Ground, and LIN. Removeable battery circuitry further includes multiple batter cells (B1-B #) wired in series. Start MOSFET Q9 is configured to block discharge current when off or disconnected from the vehicle. Charge MOSFET Q7 is configured to block charge current when cold, disconnected from the vehicle, or in the event of excessive charge voltage or current. Removeable battery circuitry includes a DC/DC converter circuit with two output voltage settings, nominally 7.5V and 12.8V, and current limiting. The current limit may change with the voltage output setting. The output of the DC/DC converter is connected to the constant voltage pin via a diode to prevent current flow when the voltage at the pin is greater than the output setting. Removeable battery circuitry further includes cell balancing circuitry and cell heating element(s). In one implementation, the cell heating element may be connected to each cell. Optionally in this implementation, the per cell heating elements are selectively turned on during charging to provide cell balancing. Additionally, removeable battery circuitry further includes current sensor R1, one or more temperature sensors, such as a thermistor, and control circuitry including a serial communications interface such as LIN.

(136) In the off state, removeable battery **1230** is configured to turn off start MOSFET Q9 and use a DC/DC converter circuitry to supply approximately 7.5V to its Constant Power pin. This voltage is sustained until the pack reaches a predetermined minimum SOC. The start field-effect transistor (FET) is configured to prevent current from flowing in case of a short circuit across two pins when the battery is not installed. The charge FET is configured to prevent charging of the battery when the cell temperature is too low. The Constant power output is current-limited to prevent damage in case of a short-circuit. The Constant power output includes a blocking diode, such that no power flows through the pin when the voltage is greater than the voltage set point. Removeable battery **1230** may receive commands over the LIN serial communications bus, such as, but not limited to, heating enable/disable, start FET on/off, and/or wake/sleep commands. Additionally, removeable battery **1230** may send status over the LIN serial communications bus, such as, but not limited to, pack voltage, current, and temperature, charge and discharge current limits, and/or heater, DC/DC converter, and start and charge FET status. For example, cell heating may be turned on when heating is enabled and the battery temperature is less than optimal for charge and discharge performance. The heaters are powered by the battery cells. As described above, in one implementation, the cell heating element may be connected to each cell. Optionally in this implementation, the per cell heating elements are selectively turned on during charging to provide cell balancing.

(137) Referring now to ISG controller circuitry of removable lithium-ion battery system **1200**, MOSFET Q1 PWMs is configured to pack voltage to supply 14V power to fuel pump **1222**. The inductance of fuel pump **1222** is utilized to filter the output of Q1. Diode D1 is configured to

provide a path for fuel pump **1222** current to flow when **Q1** is off. Diode **D7** is configured to prevent current from flowing into ISG controller **1210** when an external voltage is applied to fuel pump **1222**, such as when priming fuel pump **1222**. MOSFET **Q6** is configured to connect constant power to switched power when the key switch is closed or the power button is pressed to wake the system. Diode **D6** is configured to ensure that voltage is present at ECM **1224** when **Q6** is on, while blocking current flow when ECM output voltage is greater than the switched power output. MOSFET **Q2** is configured to be turned on to supply battery power to the three-phase bridge during starting.

(138) To start the engine, ISG controller **1210** is configured to use a flywheel encoder input and the three-phase bridge to apply variable frequency AC voltage to the stator synchronous to the flywheel position. When the engine is running, the three-phase bridge is configured to alternately shunt the stator phases and rectifies the stator output to feed the current steering circuitry. When the engine is running, MOSFETs **Q2-Q5** and diodes **D3-D5** are configured to provide current steering. At most one of MOSFET **Q2-Q5** is on at any given time, directing current to the respective output.

Capacitors **C2-C5** are configured to filter the outputs. When none of the MOSFETS are on, current flows through the body diode of **Q2** to charge **C2** and supply power to **Q1** and potentially the removeable battery. Diode **D2** is configured to supply power to the Constant Power pin, raising this voltage above the output voltage of both the removeable and fixed batteries. ISG controller **1210** includes a CAN bus serial interface to the other electronic controllers on the vehicle and a LIN bus serial interface to the removeable and fixed batteries.

(139) Referring now to FIG. **16**, a computer-implemented method **1300** for electric starting an engine from a removeable battery is shown. In the illustrative embodiment, the vehicle supports an electric start or restart of the engine while the engine is stopped by delivering electrical power to an engine control module (ECM) when a power source satisfies a predetermined characteristic. Specifically, with the key on but the engine not running, in response to a user input, the vehicle operates an integrated starter-generator (ISG) **1223** to turn the engine and the ECM operates the fuel injectors and ignition coil to start the engine.

(140) To do so, the method **1300** is performed by an integrated starter generator (ISG) controller (e.g., **1210**) of the vehicle that is communicatively coupled to the removeable battery (e.g., **1230**) to control an integrated starter-generator (ISG) (e.g., **1223**) to turn the engine. In block **1302**, while removeable battery **1230** is in the off state (e.g., **1102**), ISG controller **1210** monitors a key switch input via a key (e.g., **1236**) and/or a power button input via a power button (e.g., **1234**).

(141) In block **1304**, when the key switch is closed or the power button is pressed, ISG controller **1210** wakes removeable battery **1230** with a message on a LIN bus **1238** to provide power to a display and an engine control module (ECM) (e.g., **1224**). In some embodiments, it may also enable a removeable battery's heater.

(142) When removeable battery **1230** is in the wake state (e.g., **1104**), ISG controller **1210** determines if a power capacity of removeable battery **1230** satisfies a predetermined characteristic for the electric start of the engine, as indicated in block **1306**. As an example, the predetermined characteristic may include the power capacity of removeable battery **1240** exceeding a predetermined threshold. For example, a control logic of removeable battery **1230** may measure the battery temperature and track its state-of-charge. Removeable battery **1230** may use these measurements to periodically or continually update a maximum discharge current. Subsequently, the maximum discharge current is transmitted to ISG controller **1210** over the LIN bus **1238**.

Additionally, an engine control module (ECM) **1224** may transmit an engine coolant temperature to ISG controller **1210** over CAN. In the illustrative embodiment, ISG controller **1210** may determine the predetermined threshold for the electric start of the engine based on engine coolant temperature and time since the engine was stopped. Subsequently, ISG controller **1210** may compare the maximum discharge current received from removeable battery **1230** against the predetermined threshold for the electric start.

(143) If ISG controller **1210** determines that the power capacity does not satisfy the predetermined characteristic (e.g., below the predetermined threshold) in block **1308**, method **1300** proceeds to block **1310**. In block **1310**, ISG controller **1210** transmits a notification to a rider of the vehicle indicating that the electric start is not possible. In some embodiments, the notification may further include a message indicating that a manual start is required to start the engine. In other embodiments, the notification may include a charging status of removeable lithium-ion battery **1230**. It should be appreciated that the notification may be displayed on a gauge of the vehicle and/or a display screen that is communicatively coupled to the vehicle. Subsequently, method **1300** loops back to block **1302**.

(144) If, however, ISG controller **1210** determines that the power capacity satisfy the predetermined characteristic (e.g., above the predetermined threshold) in block **1308**, method **1300** advances to block **1312**. In block **1312**, the ISG controller **1210** transmits a notification to a rider of the vehicle indicating that the electric start is possible. For example, the notification may be displayed on a gauge of the vehicle and/or a display screen that is communicatively coupled to the vehicle.

(145) Subsequently, method **1300** proceeds to block **1314** to receive the rider input requesting to electric start. For example, the rider may use a pushbutton or a selection on a touchscreen display of the vehicle to request an engine start. In other words, ISG controller **1210** is configured to initiate an electric start when it has determined that sufficient power is available and the user then presses the start button.

(146) If ISG controller **1210** determines that a user input has not been received in block **1316**, method **1300** loops back to block **1314** to continue awaiting a user input. If, however, ISG controller **1210** determines that a user input has been received, method **1300** proceeds to block **1318**.

(147) In block **1318**, ISG controller **1210** operates the ISG **1223** to turn the engine. To do so, an electric start begins with a ramp up of the voltage on fuel pump **1222**. Once the predetermined fuel pump voltage is reached, ISG controller **1210** applies AC voltage to the stator to produce starting torque.

(148) Subsequently, the status of removeable lithium-ion battery **1230** transitions to the run state **1110** when a predetermined time has elapsed since the user input.

(149) Referring now to FIG. **19**, a computer-implemented method **1600** for starting an engine of a vehicle in reverse is shown. In the illustrative embodiment, the vehicle supports an electric start or restart of the engine in a reverse direction while the engine is stopped by delivering electrical power to an engine control module (ECM) when a power source satisfies a predetermined characteristic. Specifically, with the key on but the engine not running, in response to a user input, the vehicle operates an integrated starter-generator (ISG) to turn the engine in the reverse direction and the ECM operates the fuel injectors and ignition coil to start the engine in reverse. By going straight from engine stopped to reverse, time needed to operate the vehicle in reverse is reduced. This speeds up backing up the vehicle off a trailer or out of a shed and reduces wear and tear on the engine and starting system.

(150) To do so, the method **1600** is performed by an integrated starter generator (ISG) controller of the vehicle to control an integrated starter-generator (ISG) to turn the engine in a reverse direction. In block **1602**, the ISG controller determines if a vehicle is turned on but an engine of the vehicle is not running. In response to determining that the vehicle is turned on but the engine is not running, the ISG controller further determines if a power capacity of a power source satisfies a predetermined characteristic for the electric start of the engine, as indicated in block **1604**. For example, the predetermined characteristic may include the power capacity of the power source exceeding a predetermined threshold. The power source may include a lithium-ion battery or ultracapacitors. For example, in some embodiments, the power source may calculate and transmit its power capacity and a maximum discharge current based on its temperature and state-of-charge

to the ISG controller. The ISG controller may use the engine temperature and time stopped to determine the predetermined threshold for the electric start of the engine power capacity required to start the engine.

(151) If the ISG controller determines that the power capacity does not satisfy the predetermined characteristic (e.g., below the predetermined threshold) in block **1606**, method **1600** proceeds to block **1608**. In block **1608**, the ISG controller transmits a notification to a rider of the vehicle indicating that the electric start is not possible. In some embodiments, the notification may further include a message indicating that a manual start is required to start the engine. In other embodiments, the notification may include a charging status of the power source. Subsequently, method **1600** loops back to block **1602**.

(152) If, however, the ISG controller determines that the power capacity satisfies the predetermined characteristic (e.g., exceeds the predetermined threshold) in block **1606**, method **1600** advances to block **1610**. In block **1610**, the ISG controller transmits a notification to a rider of the vehicle indicating that the electric start is possible. For example, the notification may be displayed on a gauge of the vehicle and/or a display screen that is communicatively coupled to the vehicle.

(153) Subsequently, method **1600** proceeds to block **1612** to receive the rider input requesting to start in-reverse. For example, the rider may use a pushbutton or a selection on a touchscreen display of the vehicle to request an engine start in the reverse direction.

(154) If the ISG controller determines that a user input has not been received in block **1614**, method **1600** loops back to block **1612** to continue awaiting a user input. If, however, the ISG controller determines that a user input has been received, method **1600** proceeds to block **1616**.

(155) In block **1616**, the ISG controller operates the ISG to turn the engine in the reverse direction. To do so, the ISG controller applies voltage to a stator to produce starting torque in the reverse direction.

(156) Now referring to FIG. **20**, another computer implemented method of determining which start modes are available to an operator of vehicle **10** will be discussed. A method **2000** starts with decision block **2002** and either ISG controller **1210** or ECM **1224** determines if the available current or power capacity of battery **1230** satisfies a first predetermined characteristic to conduct an electric start of vehicle **10**. This step is similar to step **1306** of method **1300**. In the present embodiment, the first predetermined characteristic is a first current threshold. In an example thereof, in block **2002**, either of ISG controller **1210** or ECM **1224** compares the present battery current (A) to the first current threshold. In various embodiments the first predetermined characteristic may be a voltage threshold.

(157) If it is determined in decision block **2002** that the current (A) of battery **1230** is insufficient to conduct an electric start of vehicle **10**, method **2000** moves on to decision block **2018**. In decision block **2018**, ISG controller **1210** determines if the available current (A) of battery **1230** satisfies a second predetermined characteristic. In the present embodiment, the second predetermined characteristic is a second current threshold. That is, in block **2018**, either of ISG controller **1210** or ECM **1224** compares the present current (A) to the second current threshold. In the present embodiment, the second current threshold is less than the first current threshold. If it is determined that the available current (A) of battery **1230** satisfies the second predetermined characteristic, method **2000** determines that the pull assist starting method is available to the rider in block **2020**. If it is determined that the available current (A) of battery **1230** does not satisfy the second predetermined characteristic, method **2000** determines that a manual start is required by the rider in block **2022**. That is, if the available current (A) is less than the second current threshold, a notification is provided to the rider that a manual start is necessary to start the engine of vehicle **10**.

(158) If it is determined in decision block **2002** that the current (A) of battery **1230** is greater than the first current threshold, that means there is sufficient current to conduct an electric start of vehicle **10** and method **2000** moves on to decision block **2004**. In decision block **2004**, either of ISG controller **1210** or ECM **1224** determines what the current engine rotation count is and

compares that to an engine rotation count threshold. The engine rotation count is defined as the current aggregate revolutions of the crankshaft measured since the ISG controller **1210** was last in an off state. In the present embodiment, the engine rotation count threshold is four rotations. In various embodiments, the engine rotation count threshold is zero revolutions, 100 revolutions, 1000 revolutions, or another suitable number of rotations. Decision block **2004** is configured to determine if the engine of vehicle **10** is in a 'warmed-up' state or if the engine is starting from a 'cold-state' which impacts the amount of force required to start the engine.

(159) If it is determined in decision block **2004** that the engine rotation count is less than the engine rotation count threshold, method **2000** moves on to decision block **2010**. Decision block **2010** is a determination if the ISG controller **1210** is in a powered ON, or initialized state, or a powered OFF, un-initialized state. If the ECM is in a powered ON state, this is indicative that the vehicle is keyed on, and if the ECM is in a powered OFF state, this is indicative that the vehicle is keyed off. If it is determined in decision block **2010** that controller **1210** is un-initialized, method **2000** moves on to decision block **2018**, and the engine of vehicle **10** may be started in either a pull-assist mode or a manual mode, as previously described.

(160) If it is determined in decision block **2010** that controller **1210** is in an initialized state, this indicates that the rider will likely desire to turn on the engine of vehicle **10**. Subsequently, method **2000** moves on to determine an engine friction coefficient in a subprocess **2012** by conducting an engine friction coefficient test.

(161) Turning to FIG. **20A**, an exemplary engine friction coefficient test in subprocess **2012** is provided. In one embodiment, in response to a start of the engine coefficient test, ISG controller **1210** is configured to provide power to stator **118** to rotate crankshaft **460** in the reverse direction until the crankshaft stops at a first position at a first predetermined crankshaft angle. That is, in step **2030** of subprocess **2012**, crankshaft **460** is rotated, or swept, using the ISG **1223**. In the present embodiment, the first position is a top dead center (TDC) position when the piston is in the compression stroke. When crankshaft **460** is in the first position, ISG controller **1210** provides power to stator **118** to rotate crankshaft **460** in the forward direction to a second position at a second predetermined crankshaft angle. That is, ISG **1223** sweeps the crankshaft **460** from the first position to the second position. As illustrated in block **2032**, either of, or both of, ISG controller **1210** or ECM **1224** is configured to measure a representative crankshaft movement characteristic while crankshaft is moving between the first position and the second position. In various embodiments, the first position and the second position are defined as the crankshaft position when the ports (e.g., intake, exhaust, and/or transfer ports) are closed. That is, in a two-cylinder, two-stroke engine, the crankshaft **460** is rotated in a first direction to the first position so that a first piston within the first cylinder of the two cylinders is translated to a position where the exhaust port and transfer port of the first cylinder are closed, trapping air within the first cylinder and building compression within the first cylinder. Subsequently, crankshaft **460** is rotated in a second direction, opposite the first direction, to the second position so that a second piston within the second cylinder of the two cylinders is translated to a position where the exhaust port and transfer port of the second cylinder are closed, trapping air within the second cylinder and building compression within the second cylinder. In a four-stroke engine, both intake and exhaust valves would need to be closed during the crankshaft rotation between the first position and second position to build up compression within the cylinder.

(162) In the present embodiment, either of ISG controller **1210** or ECM **1224** is configured to measure a maximum RPM (RPM_max). In various embodiments, the measured crankshaft movement characteristic is an acceleration value or an average RPM (RPM_avg). As represented in block **2036**, either of, or both of, ISG controller **1210** or ECM **1224** is configured to determine the amount of torque provided to the crankshaft **460** by stator **118**. In the present embodiment, the amount of torque provided to the crankshaft **460** is determined by the amount of power input to the stator **118**. Subsequently in subprocess **2012**, ISG controller **1210** uses the determined maximum

RPM of the crankshaft **460** and the determined amount of torque provided to the crankshaft **460** to determine an engine friction coefficient (efc). In an example thereof, the engine friction coefficient is defined by Equation 1: $efc = \text{torque} * (100 / \text{RPM_max})$.

(163) In another embodiment, ISG controller **1210** provides instructions and a brief pulse of power to stator **118** to rotate crankshaft **460** within the engine. In the present embodiment, crankshaft **460** is configured to rotate between compression strokes of a 2-stroke engine, or sweep, between compression strokes. In various embodiments, crankshaft **460** is configured to sweep between +90/-90 degrees from center and/or the crankshaft starting point. In various embodiments, crankshaft **460** is configured to sweep between +120/-120 degrees from center and/or the crankshaft starting point. In various embodiments, crankshaft **460** is configured to sweep between +45/-45 degrees from center and/or the crankshaft starting point. In various embodiments, the crankshaft is configured to sweep as far as it is able to under a predetermined amount of power provided to the stator over a predetermined amount of time. In one example, ISG controller **1210** provides 100 W for one second to stator and ISG controller **1210** measures the total rotation achieved by crankshaft **460**. In the present embodiment, ISG controller **1210** would determine the torque provided to the crankshaft **460** based upon the power provided to the stator **118** as well as determine the measured average rotation rate (RPM_avg) of the crankshaft based upon the total rotation achieved over the time power was provided to the stator **118**. In an example thereof, the engine friction coefficient is defined by Equation 2: $efc = \text{torque} * (100 / \text{RPM_avg})$.

(164) In the present embodiment, ISG controller **1210** uses both of (1) the amount of power generated and/or provided to the stator **118**; and (2) the measured movement characteristic of the crankshaft **460** to determine an engine friction coefficient. In other words, the ISG controller **1210** conducts the engine friction coefficient test on the crankshaft **460** to determine how difficult it is to rotate the crankshaft **460** and start the engine.

(165) Referring again to FIG. **20**, after the engine friction coefficient (efc) is determined in step **2012**, method **2000** proceeds to decision block **2014**. In decision block **2014** either of ISG controller **1210** or ECM **1224** compares the determined engine friction coefficient to a predetermined engine friction coefficient threshold, referred to in FIG. **20** as a start friction threshold. If it is determined, in decision block **2014**, that the determined engine friction coefficient is less than the predetermined engine friction coefficient threshold, this indicates that the frictional forces present in the engine are low enough for the electric start mode to be an effective method to start the engine of vehicle **10**. If it is determined that the engine friction coefficient is less than the predetermined engine friction coefficient threshold, method **2000** moves on to step **2016**. In step **2016**, method **2000** is configured to enable the electric start mode, as previously described, and provide an indication or notification to the rider that the electric start mode is available to the rider.

(166) If it is determined in step **2012** that the determined engine friction coefficient is greater than the predetermined engine friction coefficient threshold, this indicates that the electric start may be ineffective or inefficient at starting the engine of vehicle **10**. That is, if the electric start mode is used when the determined engine friction coefficient is greater than the predetermined engine friction coefficient, the engine of vehicle **10** may stall out during the startup sequence or may be unable to sufficiently rotate the crankshaft **460** to start or support the ignition cycle. If it is determined in step **2012** that the determined engine friction coefficient is greater than the predetermined engine friction coefficient threshold, method **2000** moves on to decision block **2018**, where vehicle **10** can be started in either a pull-assist mode or a manual mode, as previously described.

(167) Still referring to FIG. **20**, if it is determined in decision block **2004** that the engine rotation count is greater than the engine rotation count threshold, this may be indicative that the vehicle was previously running, and method **2000** moves on to decision block **2006**. In decision block **2006**, it is determined if ISG controller **1210** is in a standby mode. In examples thereof, standby mode may be when the controller **1210** is turned on, or initialized, but the engine of vehicle **10** is off. If the

controller is not in standby mode, this may be indicative that the controller is also powered off, or uninitialized. If it is determined in block **2006** that the controller is not in standby mode (e.g., the controller is powered off) method **2000** is completed. If it is determined in block **2006** that the controller **1210** is in standby mode, process **2000** proceeds to decision block **2008** to determine if a start button has been actuated. In various embodiments, decision block **2008** determines if a user input has been received indicating a desire to start the engine of vehicle **10**. In various embodiments, the user input could be an input to a tactile button, a switch, a slider, a user interface, a display, such as on the vehicle or on a wearable accessory, or the like. If it is determined in decision block **2008** that a start button has not been actuated, method **2000** is completed. If it is determined in decision block **2008** that a start button has been actuated, this is an indication of a desire to start the engine of vehicle **10**, and method **2000** proceeds to subprocess **2012** and the engine friction coefficient test will be conducted, as previously explained.

(168) Referring now to FIG. **21**, an alternate method **2040** of determining a start mode for the engine of vehicle **10** is provided. Method **2040** is similar to method **2000** in several ways. Method **2040** starts either at block **2041** if controller **1210** is in a motor mode or a generator mode or at block **2042** if controller **1210** is in standby mode. In the present embodiment, motor mode is indicative of a state where controller **1210** is driving current into stator **118** to turn crankshaft **460**. Further, generator mode is indicative of a state where controller **1210** is pulling current from stator **118** to provide power to other electrical loads on vehicle **10**.

(169) If method **2040** starts in block **2041**, the first step is decision block **2002** and either ISG controller **1210** or ECM **1224** determines if the available current or power capacity of battery **1230** satisfies a first predetermined characteristic to conduct an electric start of vehicle **10**. This step is similar to step **1306** of method **1300**. In the present embodiment, the first predetermined characteristic is a first current threshold. In an example thereof, in block **2002**, either of ISG controller **1210** or ECM **1224** compares the present battery current (A) to the first current threshold. In various embodiments the first predetermined characteristic may be a voltage threshold.

(170) If it is determined in decision block **2002** that the current (A) of battery **1230** is insufficient to conduct an electric start of vehicle **10**, method **2040** moves on to decision block **2018**. In decision block **2018**, ISG controller **1210** determines if the available current (A) of battery **1230** satisfies a second predetermined characteristic. In the present embodiment, the second predetermined characteristic is a second current threshold. That is, in block **2018**, either of ISG controller **1210** or ECM **1224** compares the present current (A) to the second current threshold. In the present embodiment, the second current threshold is less than the first current threshold. If it is determined that the available current (A) of battery **1230** satisfies the second predetermined characteristic, method **2040** determines that the pull assist starting method is available to the rider in block **2020**. If it is determined that the available current (A) of battery **1230** does not satisfy the second predetermined characteristic, method **2040** determines that a manual start is required by the rider in block **2022**. That is, if the available current (A) is less than the second current threshold, a notification is provided to the rider that a manual start is necessary to start the engine of vehicle **10**.

(171) If it is determined in decision block **2002** that the current (A) of battery **1230** is greater than the first current threshold, that means there is sufficient current to conduct an electric start of vehicle **10** and method **2000** moves on to decision block **2010**. Decision block **2010** is a determination if the ISG controller **1210** is in a powered ON state, or initialized, or a powered OFF state, or un-initialized. If the ECM is in a powered ON state, this is indicative that the vehicle is keyed on, and if the ECM is in a powered OFF state, this is indicative that the vehicle is keyed off. If it is determined in decision block **2010** that controller **1210** is un-initialized, method **2040** moves on to decision block **2018**, and the engine of vehicle **10** may be started in either a pull-assist mode or a manual mode, as previously described.

(172) If it is determined in decision block **2010** that controller **1210** is in an initialized state, this

indicates a desire by the rider to turn on the engine of vehicle **10** and subsequently method **2040** moves on to determine an engine friction coefficient in subprocess **2012**, as previously described and shown in FIG. **20A**.

(173) Referring still to FIG. **21**, after the engine friction coefficient (efc) is determined in step **2012**, method **2040** proceeds to decision block **2014**. In decision block **2014** either of ISG controller **1210** or ECM **1224** compares the determined engine friction coefficient to a predetermined engine friction coefficient threshold, referred to in FIG. **21** as a start friction threshold. If it is determined, in decision block **2014**, that the determined engine friction coefficient is less than the predetermined engine friction coefficient threshold, this indicates that the frictional forces present in the engine are low enough for the electric start mode to be an effective method to start the engine of vehicle **10**. If it is determined that the engine friction coefficient is less than the predetermined engine friction coefficient threshold, method **2040** moves on to step **2016**. In step **2016**, method **2040** is configured to enable the electric start mode, as previously described, and provide an indication or notification to the rider that the electric start mode is available to the rider.

(174) If it is determined in step **2012** that the determined engine friction coefficient is greater than the predetermined engine friction coefficient threshold, this indicates that the electric start may be ineffective or inefficient at starting the engine of vehicle **10**. That is, if the electric start mode is used when the determined engine friction coefficient is greater than the predetermined engine friction coefficient, the engine of vehicle **10** may stall out during the startup sequence or may be unable to sufficiently rotate the crankshaft **460** to start or support the ignition cycle. If it is determined in step **2012** that the determined engine friction coefficient is greater than the predetermined engine friction coefficient threshold, method **2040** moves on to decision block **2018**, where vehicle **10** can be started in either a pull-assist mode or a manual mode, as previously described.

(175) Still referring to FIG. **21**, if controller **1210** is in standby mode, method **2040** will start in block **2042**. The first step when controller **1210** is in standby mode is to monitor for an actuation of a start button, as previously described in block **2008**. If no start button actuation is received, method **2040** is completed. If a start button is actuated, this is indicative of a desire by the rider to start the engine of vehicle **10**, and method **2040** will proceed to complete the engine friction coefficient test in subprocess **2012**.

(176) Turning now to FIG. **22**, a method **2050** will be described which is configured to determine in which operating mode the engine of vehicle **10** is enabled. In the present embodiment, method **2050** starts with a determination, in block **2052**, if the controller **1210**, **1224** is in a motor mode. In motor mode, controller **1210**, **1224** is configured to drive current into stator **118** to turn the crankshaft **460**, as previously described. If it is determined, in decision block **2052**, that controller **1210**, **1224** is not in motor mode, but in another mode, method **2050** proceeds to block **2054**. In decision block **2054**, either of ECM **1224** or ISG controller **1210** measures the engine speed of the engine of vehicle **10** at the crankshaft **460** or flywheel. More specifically, decision block **2054** determines if a current engine speed of the engine is equal to zero. If it is determined that the engine speed is equal to zero, method **2050** proceeds to block **2058** and controller **1210**, **1224** enters into, or remains in, standby mode and controller **1210** remains in an initiated state but the engine is not running. If it is determined that the engine speed is not equal to zero, method **2050** proceeds to block **2056** and controller **1210**, **1224** enters into, or remains in, a generator mode and controller **1210** is configured to pull power from stator **118** to provide power to other vehicle electrical loads. Block **2054** is configured to determine whether or not the engine of vehicle **10** is currently running, thereby indicating a desired use case of the engine, and stator **118**, of vehicle **10**.

(177) If it is determined in decision block **2052** that the controller is in motor mode, method **2050** proceeds to decision block **2060**. Decision block **2060** is determined to monitor and compare the current number of revolutions experienced by crankshaft **460** during a starting attempt to a revolutions threshold. Because motor mode is configured to enable controller **1210** to drive current

into stator **118** to turn crankshaft **460**, decision block **2060** will be queried during a start attempt of the engine. If it is determined in decision block **2060** that the crankshaft **460** has rotated a greater number of revolutions than the revolutions threshold, method **2050** proceeds to block **2064**, and motor mode is disabled and standby mode is set. This indicates the end of an attempt to start the engine of vehicle **10**. A rider may subsequently attempt to start the engine again any number of times. In the present embodiment, the revolutions threshold is four revolutions. In various embodiments, the revolutions threshold may be six revolutions, eight revolutions, or more revolutions. In various embodiments, the revolutions threshold may be based temporally, and the threshold may be one second, two seconds, three seconds, four seconds, or another time period. If it is determined in decision block **2060** that the crankshaft **460** has rotated a fewer number of revolutions than the revolutions threshold, method **2050** proceeds to block **2062**.

(178) In decision block **2062**, a PID controller is utilized to create a closed loop feedback circuit to monitor the current going into the stator **118** from the ISG controller **1210**. The PID controller receives a quadrature current (A) value of the battery (e.g., battery **1230**) being used to drive current into the stator **118** and compares that to a reference current (A) value from the stator **118**. Before attempting to start the engine, the error will be zero. As stator **118** rotates crankshaft **460** and the engine starts running, the error between the current (A) provided to the stator (**118**) and the reference current (A) value from the stator **118** will start to deviate because the stator will start to be rotated by the operation of the engine. During the startup of the engine, the error will continue to rise until the stator is producing enough power to overcome the voltage of the battery (e.g., battery **1230**). When it is determined that the absolute value of the error between the quadrature current (A) and the reference current (A) from the stator **118** has increased to be greater than an error threshold, method **2050** proceeds to block **2064** and motor mode is disabled and controller **1210** enters standby mode. If the error between the quadrature current (A) and the reference current (A) reaches a value greater than the error threshold, this is indicative that the engine is now running and starting to create more current out of stator **118**.

(179) Block **2062** is configured to protect the controller **1210** and battery (e.g., battery **1230**) from an uncontrolled charge current surging backwards from stator **118** through controller **1210** and battery (e.g., battery **1230**). If it is determined that the absolute value of the error between the quadrature current (A) and the reference current (A) from the stator **118** has is still less than the error threshold, method **2050** proceeds to block **2066** and controller **1210** remains in motor mode. Subsequently, method **2050** is completed and may be repeated.

(180) Both comparisons completed in decision blocks **2060** and **2062** are designed to protect the engine and battery (e.g., battery **1230**). That is, decision block **2060** is configured to limit the number of revolutions in a given start attempt, and decision block **2062** is configured to determine when the engine is running to prevent uncontrolled charge currents from flowing to the controller **1210** or battery (e.g., **1230**).

(181) Now referring to FIGS. **23-24**, vehicle **10** may be configured with a shunt **19** configured to shunt the stator **118** to a ground terminal, or grounded pin **119**. In the present embodiment, a method **2100** is provided which determines when stator **118** may be shunted to the grounded pin **119**. In method **2100**, shunting stator **118** to grounded pin **119** substantially removes the resistance and/or drag that alternator **20** puts on the engine of vehicle **10**, thereby increasing the total power output of the engine to the ground engaging members of vehicle **10**. Method **2100** starts with a decision block **2102** where either of ISG controller **1210** or ECM **1224** determines if a user or operator of vehicle **10** has provided an input to request a higher amount of power. In the present embodiment, a request for a high power input may be a request to change drive modes, or performance modes, such as for a Sport Mode, a Track Mode. In various embodiments, as another high power input, vehicle **10** may comprise a Boost button which may trigger an acceleration event or promote acceleration by vehicle **10**. In various embodiments, decision block **1210** may be configured to determine if vehicle **10** is in an ECO mode, or other mode configured to conserve

fuel or otherwise operate the engine more economically. That is, by shunting stator **118** to a ground terminal, crankshaft **460** is able to move with less resistance, and provide more power to the ground engaging members and/or use less fuel during normal operation.

(182) If it is determined in decision block **2102** that a high power request has not been requested by a user of the vehicle **10**, method **2100** proceeds to decision block **2104**. In decision block **2104**, either of ISG controller **1210** or ECM **1224** is configured to determine if a high power event has been detected. In the present embodiment, a high power event is an acceleration event, a towing event, or another type of higher power event. A high power event may be detected by an engine speed sensor, a throttle valve position sensor, a throttle position sensor, a shock position sensor, a GPS, an IMU, an accelerometer, a gyroscope, or another type of vehicle condition sensor supported by vehicle **10**.

(183) If it is determined in decision block **2104** that a high power request is not detected, method **2100** proceeds back to the start and proceeds through decision block **2102** again. If it is determined in decision block **2104** that a high power request is detected, method **2100** proceeds to decision block **2106**. Further, if it is determined in decision block **2102** that a high power request has been requested by a user of vehicle **10**, method **2100** proceeds to decision block **2106**. In decision block **2106**, either ISG controller **1210** or ECM **1224** is configured to compare the current voltage (V) or available current (A) level of battery **18** to a first current threshold. In the present embodiment, decision block **2106** is configured to determine if the battery **18** can sufficiently provide power (based upon the current voltage (V) or available current (A) level of battery **18**) to the various electrical systems of vehicle **10** without battery **18** being presently charged by alternator **20**. In the present embodiment, the first threshold is 24V. In various embodiments, the first threshold is 12V, 14V, 16V, or another voltage. In various embodiments, the first threshold is a critical voltage level based upon the size of battery **18**. If it is determined that the current voltage (V) or current (A) level of battery **18** is less than the first threshold, method **2100** proceeds to block **2107** and a notification or indication is provided to the user of vehicle **10** that shunting stator **18** to the grounded pin **119** is unavailable.

(184) If it is determined in decision block **2106** that the current voltage (V) or current (A) level, of battery **18** is greater than the first threshold, method **2100** moves on to decision block **2108**. In decision block **2108**, either of ISG controller **1210** or ECM **1224** are configured to compare the current voltage (V) or current (A) level of battery **18** to a second threshold. In the present embodiment, the second threshold is greater than the first threshold. In the present embodiment, the second threshold is 28V. If it is determined that the current voltage (V) or current (A) level of battery **18** is less than the second threshold, method **2100** proceeds to block **2110** and either of controller **1210**, **1224** is configured to shed other voltage loads on vehicle **10**. In various embodiments controller **1210**, **1224** is configured to shed voltage loads based upon criticality, voltage level, or a user selected order. In the present embodiment, voltage loads are shed until the current voltage (V) or current (A) level of battery **18** is greater than the second threshold. In various embodiments, a group of components, such as a chassis group of components (e.g., heaters, speakers, etc.) are shed together in block **2110**. Additional details regarding load shedding of components may be found in U.S. application Ser. No. 16/560,588, filed Jun. 25, 2020, titled MANAGING RECREATIONAL VEHICLES AND ACCESSORIES, the entire disclosure of which is expressly incorporated herein by reference.

(185) If it is determined, in decision block **2108**, that the current voltage (V) or current (A) level of battery **18** is greater than the second threshold, method **2100** proceeds to block **2112**. Additionally, if vehicle voltage loads have been shed in block **2110**, method **2100** proceeds from block **2110** to block **2112**. In block **2112**, method **2100** completes the shunting process, and shunts stator **118** to the grounded pin **119**, providing a path of least resistance between the stator **118** and the grounded pin **119** and crankshaft **460** may rotate with less resistance. In various embodiments, shunting stator **118** to the grounded pin **119** adds an additional 0.1 kW-3 kW of power to the vehicle **10**. That is,

shunting stator **118** to the grounded pin **119** removes 0.1 kW-3 kW of resistance-based losses from the engine. In various embodiments, shunting stator **118** to the grounded pin effectively adds more than 3 kW of additional power to the vehicle **10**. In various embodiments, an indication is provided to the rider of vehicle **10** of how much additional power is being output as a result of shunting stator **118** to the grounded pin **119**. The indication may be a displayed value on a user interface, a mobile device, or another visual component.

(186) The above specification, examples and data provide a complete description of the manufacture and use of the composition of the invention. Since many embodiments of the invention can be made without departing from the spirit and scope of the invention, the invention resides in the claims hereinafter appended.

Claims

1. A method for controlling different starting modes of an internal combustion engine of a vehicle, the method comprising: (a) determining, by a controller, a power level of a power source of the vehicle; (b) determining, by the controller, that the power level of the power source is above a first predetermined threshold, and when the controller determines that the power level is above the first predetermined threshold, the controller also determining that the power level is above a second predetermined threshold and determining that an engine friction coefficient is above a start friction threshold, wherein the engine friction coefficient is determined by the controller based on a determined maximum revolutions per minute of a crankshaft of the internal combustion engine and a determined amount of torque provided to the crankshaft; (c) determining, by the controller, that a pull-assist starting mode for the internal combustion engine is available in response to the power level of the power source being below above the first predetermined threshold and above the second predetermined threshold and the engine friction coefficient being above the start friction threshold; (d) transmitting a notification to a user indicating an available starting mode, the available starting mode being the pull-assist starting mode; (e) receiving a user request to start the internal combustion engine subsequent to determining the power level of the power source and transmitting the notification to the user indicating the available starting mode is the pull-assist starting mode, wherein the user request is received via a user input sensor of the vehicle; and (f) applying an assist torque to the internal combustion engine in response to receiving the user request, wherein steps (a)-(f) are performed sequentially.
2. The method of claim 1, wherein the user input sensor of the vehicle is selected from the group consisting of a button, a pull recoil handle sensor, a pull force sensor, and/or a recoil rotation sensor.
3. The method of claim 1, wherein the pull-assist starting mode supports electric start with manual pull assist.
4. The method of claim 1, wherein in the pull-assist starting mode, when a rope of a recoil starter is pulled by the user, current is driven into a stator to apply cranking assist torque to the crankshaft to turn the internal combustion engine.
5. The method of claim 4, wherein the current is provided by a battery system.
6. The method of claim 5, wherein the battery system includes a battery and a DC/DC converter.
7. The method of claim 6, wherein the battery system further includes a capacitor system.
8. The method of claim 1, wherein the notification is displayed on a display screen of the vehicle.
9. The method of claim 1, wherein the notification is a text notification and/or a light indicator with a blinking pattern and/or a color that corresponds to the available starting mode.
10. The method of claim 1, wherein the user input sensor is positioned separate from a pull recoil starter.
11. A system on a vehicle that controls different starting modes of an internal combustion engine of the vehicle, the system comprising: (a) a controller that is configured to determine a power level of

a power source of the vehicle; (b) the controller is configured to determine whether the power level of the power source is above or below a first predetermined threshold; (i) if the controller determines that the power level is below the first predetermined threshold, the controller is configured to determine whether the power level is above a second predetermined threshold, and (ii) if the controller determines that the power level is above the first predetermined threshold, the controller is configured to determine whether the power level is above the second predetermined threshold and also determine whether an engine friction coefficient is above a start friction threshold, wherein the engine friction coefficient is determined by the controller based on a determined maximum revolutions per minute of a crankshaft of the internal combustion engine and a determined amount of torque provided to the crankshaft; (c) the controller is configured to determine that a pull-assist starting mode for the internal combustion engine is available in response to the power level of the power source being below the first predetermined threshold and above the second predetermined threshold; (d) the controller is configured to determine that the pull-assist starting mode for the internal combustion engine is available in response to the power level of the power source being above the first predetermined threshold and above the second predetermined threshold and the engine friction coefficient being above the start friction threshold; (e) the controller is configured to transmit a notification to a user indicating that the pull-assist starting mode is available; (f) the controller is configured to receive a user request to start the internal combustion engine subsequent to determining the power level of the power source and transmitting the notification to the user indicating that the pull-assist starting mode is available, wherein the user request is received via a user input sensor of the vehicle; and (g) the controller is configured to apply an assist torque to the internal combustion engine in response to receiving the user request.

12. A vehicle comprising: an internal combustion engine having a crankshaft; a power source that provides power for the vehicle; a user input sensor; and a controller controlling operation of the vehicle, wherein the controller is: (a) configured to determine a power level of the power source of the vehicle; (b) configured to determine whether the power level of the power source is above or below a first predetermined threshold; (i) if the controller determines that the power level is below the first predetermined threshold, the controller is configured to determine whether the power level is above a second predetermined threshold, and (ii) if the controller determines that the power level is above the first predetermined threshold, the controller is configured to determine whether the power level is above the second predetermined threshold and also determine whether an engine friction coefficient is above a start friction threshold, wherein the engine friction coefficient is determined by the controller based on a determined maximum revolutions per minute of the crankshaft of the internal combustion engine and a determined amount of torque provided to the crankshaft; (c) the controller is configured to determine that a pull-assist starting mode for the internal combustion engine is available in response to the power level of the power source being below the first predetermined threshold and above the second predetermined threshold; (d) the controller is configured to determine that the pull-assist starting mode for the internal combustion engine is available in response to the power level of the power source being above the first predetermined threshold and above the second predetermined threshold and the engine friction coefficient being above the start friction threshold; (e) the controller is configured to transmit a notification to a user indicating that the pull-assist starting mode is available; (f) the controller is configured to receive a user request to start the internal combustion engine subsequent to determining the power level of the power source and transmitting the notification to the user indicating that the pull-assist starting mode is available, wherein the user request is received via the user input sensor of the vehicle; and (g) the controller is configured to apply an assist torque to the internal combustion engine in response to receiving the user request.

13. The vehicle of claim 12, wherein the vehicle comprises a snowmobile, an all-terrain vehicle, a motorcycle, a watercraft, a utility vehicle, or a golf cart.

